

MAHLER'S PROBLEM ON LIOUVILLE NUMBERS

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ABSTRACT. A classical theorem of Maillet states that every nonconstant rational function with rational coefficients maps Liouville numbers to Liouville numbers. In 1984, Mahler asked whether there exists a transcendental entire function with the same property. We resolve this question in the negative. More generally, we prove that every real-analytic function on an interval with this property is the restriction of a rational function in $\mathbb{R}(x)$. The local result is quantitative: for every nonrational real-analytic function, every nonempty open subinterval of its domain contains a Liouville number whose image has irrationality exponent at most 100. The proof rests on a two-height counting estimate that treats source and target denominators independently. An adaptive determinant argument, uniform Wronskian sublevel bounds, and Farey separation establish the estimate; a nested-interval construction yields the quantitative result.

1. INTRODUCTION

A real number ξ is called a *Liouville number* if, for every integer $N \geq 1$, there exist infinitely many pairs $(p, q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$, with $q \geq 2$, such that

$$(1.1) \quad 0 < \left| \xi - \frac{p}{q} \right| < q^{-N}.$$

We denote the set of Liouville numbers by $\mathcal{L}$. Since Liouville introduced these numbers [18, 19], their exceptional rational approximation properties have made them a natural setting for the study of arithmetic properties of functions; see, for instance, [10, 12, 35].

A classical result of Maillet [22] states that every nonconstant rational function $R \in \mathbb{Q}(x)$ maps $\mathcal{L}$ into itself. We say that a function has *Maillet's property* if it preserves every Liouville number in its domain.

In one of his final papers, published in 1984, Mahler [21] posed eight problems at the intersection of transcendence theory and Diophantine approximation. The first, motivated by Maillet's theorem, asks:

Which analytic functions $f(z)$ have the property that if ξ is any Liouville number, then so is $f(\xi)$? In particular, are there entire transcendental functions with this property?

2020 *Mathematics Subject Classification.* Primary 11J82; Secondary 11J81, 26E05, 30D20.

Key words and phrases. Liouville numbers, Mahler's problem, Maillet's property, real-analytic functions, two-height approximation, arithmetic rigidity, irrationality exponent.

Supported by CNPq, Grant No. 304467/2023-5.

Mahler thus asked whether Maillet's arithmetic invariance can persist for a transcendental entire function. Subsequently recorded as an open problem by Waldschmidt [36, Question 3.27, p. 281] and Bugeaud [10, Problem 44, p. 215], this question has remained open for over four decades.

We answer Mahler's question in the negative.

Theorem 1.1. *Let $F : \mathbb{C} \rightarrow \mathbb{C}$ be entire. If*

$$F(\mathcal{L}) \subseteq \mathcal{L},$$

then $F \in \mathbb{R}[z]$. In particular, no transcendental entire function has Maillet's property.

The requirement that *every* Liouville number be preserved is essential in Theorem 1.1. Classical arithmetic interpolation allows considerable flexibility on countable sets [30, 31, 37]: transcendental entire functions can even preserve every algebraic number field [33]; see also [9, 20, 32]. Preservation of Liouville numbers on large sets is likewise compatible with much greater flexibility. Alniaçik and Saias [1] showed that a continuous, nowhere locally constant real function maps a dense G_δ subset of $\mathcal{L}$ in its domain into $\mathcal{L}$, and transcendental entire functions can preserve distinguished subclasses of $\mathcal{L}$ [23, 24].

The regularity boundary is sharp within the classical smooth–analytic hierarchy. The flexibility established at finite orders of smoothness [17] persists in the C^∞ category: there exist transcendental C^∞ diffeomorphisms of $\mathbb{R}$, arbitrarily close to the identity in the compact-open C^∞ topology, such that every derivative of every positive iterate has Maillet's property [25]. By contrast, Corollary 1.3 shows that real analyticity forces rational rigidity.

Theorem 1.1 follows from a local quantitative result. Recall that the irrationality exponent of $y \in \mathbb{R}$ is

$$(1.2) \quad \mu(y) := \sup \left\{ \lambda > 0 : \begin{array}{l} 0 < |y - a/b| < b^{-\lambda} \\ \text{for infinitely many } (a, b) \in \mathbb{Z} \times \mathbb{Z}_{>0} \end{array} \right\}.$$

Thus

$$y \in \mathcal{L} \iff \mu(y) = +\infty.$$

Theorem 1.2. *There exists an absolute constant $\tau > 2$ with the following property. Let $U \subseteq \mathbb{R}$ be an open interval and let $f : U \rightarrow \mathbb{R}$ be real-analytic. Suppose that f is not the restriction to U of a rational function in $\mathbb{R}(x)$ without poles on U . Then every nonempty open subinterval $V \subseteq U$ contains a point $\xi \in \mathcal{L}$ such that*

$$\mu(f(\xi)) \leq \tau.$$

In fact, ξ may be chosen so that

$$\left| f(\xi) - \frac{a}{b} \right| > b^{-\tau}$$

for every $a \in \mathbb{Z}$ and every sufficiently large integer b . One may take $\tau = 100$.

The particular choice $\tau = 100$ is not intrinsic and has not been optimized. The substantive point is the existence of an absolute finite bound, independent of the function. Taking the contrapositive of Theorem 1.2 gives the corresponding local rigidity statement.

Corollary 1.3. *Let $f : U \rightarrow \mathbb{R}$ be real-analytic on an open interval. If*

$$f(\mathcal{L} \cap U) \subseteq \mathcal{L},$$

then there exists a rational function $R \in \mathbb{R}(x)$, with no pole in U , such that

$$f(x) = R(x) \quad (x \in U).$$

The rational-rigidity conclusion raises a natural arithmetic question: must the resulting rational function be defined over $\mathbb{Q}$? The answer displays a sharp distinction between degree one and higher degree.

Theorem 1.4. *Let*

$$M(x) = \frac{ax + b}{cx + d} \in \mathbb{R}(x), \quad ad - bc \neq 0.$$

Then M has Maillet's property if and only if $M \in \mathbb{Q}(x)$. Moreover, if $M \notin \mathbb{Q}(x)$, then every nonempty open interval contained in the domain of M contains a point $\xi \in \mathcal{L}$ such that $M(\xi) \notin \mathcal{L}$.

This rigidity already fails for monomials of every higher degree.

Theorem 1.5. *There exist continuum many positive transcendental real numbers σ with the following property: for every integer $m \geq 2$ and every $\xi \in \mathcal{L}$,*

$$\sigma \xi^m \in \mathcal{L}.$$

Thus coefficient rigidity holds for rational maps of degree one but fails in every degree at least two. In particular, the conclusion $F \in \mathbb{R}[z]$ in Theorem 1.1 cannot be strengthened to $F \in \overline{\mathbb{Q}}[z]$. We do not attempt here to classify all rational functions in $\mathbb{R}(x)$ with Maillet's property.

The passage to the entire case is immediate. An entire function preserving $\mathcal{L}$ is real-valued on $\mathbb{R}$ by density and continuity. Corollary 1.3 makes its restriction rational; analytic continuation and the absence of poles then force it to be a polynomial.

The two-height mechanism. The local theorem requires arbitrarily good rational approximation in the source and exclusion of very good rational approximation in the image at every sufficiently large denominator. We construct such a point by counting source fractions p/q , with $q \asymp Q$, whose images are close to target fractions a/b , keeping the two denominator scales separate.

Small target denominators are handled by elementary counting. For larger denominators, write $b \asymp Q^u$ and adapt the source degree to u , while keeping

the target degree equal to one. A determinant comparison places the rational pairs in each short source cell on a common relation

$$A(x) + B(x)y = 0.$$

Such a relation cannot hold identically on the graph of a nonrational f ; this is why the argument also applies to nonrational algebraic functions. After localization away from the relevant Wronskian zeros, sublevel estimates uniform in the normalized coefficients control both the length and the number of components of the set where $A + Bf$ is small. Farey separation then gives a bounded number of source fractions per cell, while the choice of degree keeps the number of cells subquadratic in Q .

The resulting estimate has a quadratic term controlled by the lower target cutoff and a subquadratic remainder uniform in that cutoff. The constant multiplying the quadratic term is independent of the source subinterval. These uniformities allow safe rational centers to be chosen repeatedly on shrinking intervals.

We then choose nested intervals near these centers. The source approximation orders tend to infinity, and target avoidance is imposed with a margin that persists throughout each interval. Consecutive target-denominator ranges cover all sufficiently large denominators. The limiting point is therefore Liouville, while its image has irrationality exponent at most 100. Liouville numbers enter only in this final construction; the counting estimate concerns rational approximation to the graph itself.

The determinant argument lies in the tradition of Bombieri–Pila and Pila [7, 27, 28]; related developments include [3, 11, 13, 14, 15, 16, 29, 34]. Separate coordinate denominators and Wronskian arguments already occur in Pila’s work.

The results of Section 7 are independent of the analytic counting argument. The degree-one classification uses a Diophantine dichotomy for the coefficient vector and rational reconstruction from three points. The higher-degree examples combine a continued-fraction construction, Petruska’s theorem on strong Liouville numbers, and a gap principle for U_m -numbers.

A formalization of Theorem 1.2 in the Lean 4 proof assistant is available at <https://github.com/DiegoMarquesMath/MahlerLean>.

Acknowledgments. The author is grateful to Yuri Bilu, Carlos Gustavo Moreira, Johannes Schleisitz, and Jean Lelis for many valuable discussions related to this problem over the years. The author is especially grateful to Michel Waldschmidt, who first introduced him to the problem at the 2008 Arizona Winter School. He also thanks Yann Bugeaud for his monograph *Approximation by Algebraic Numbers*, which later brought the problem again to his attention.

2. RATIONAL COUNTING PRELIMINARIES

Notation and conventions. For a finite union of intervals $E \subseteq \mathbb{R}$, we denote its Lebesgue measure by $|E|$. We write $J \Subset U$ if $\bar{J}$ is a compact subset of U .

The symbols $O(\cdot)$, $\ll$, and $\asymp$ have their usual meanings. Unless otherwise stated, implied constants may depend on quantities fixed before the estimate, but not on parameters subsequently allowed to vary or tend to infinity. Additional dependence is indicated by subscripts; thus $X \ll_{A,J} Y$ means that $X \leq C(A, J)Y$ for some $C(A, J) > 0$. Generic constants $C, C_1, C_2, \dots$ may vary from one occurrence to the next.

Subscripts and superscripts on named constants are mnemonic labels, not indicators of parameter dependence; for example, ‘sub’ and ‘low’ refer to sublevel and low-target-height estimates, respectively.

With these conventions in place, we record the two rational-counting estimates needed in Section 5. They provide complementary upper and lower bounds for reduced rational points of comparable height. The first is based on the separation of reduced fractions; the absence of a factor Q in the component term will be essential in Section 5.

Lemma 2.1 (Reduced fractions in a finite union of intervals). *Let $E \subseteq \mathbb{R}$ be a union of at most R intervals, where $R \in \mathbb{Z}_{>0}$, and let $Q \in \mathbb{Z}_{>0}$. Reduced fractions are counted as rational numbers, so their positive-denominator representations are unique. Then*

$$(2.1) \quad \# \left\{ \frac{p}{q} \in E : \gcd(p, q) = 1, Q \leq q < 2Q \right\} \leq 4Q^2|E| + R.$$

Proof. Write E as the disjoint union of its nonempty connected components $I_1, \dots, I_s$, where $s \leq R$. If $p/q \neq p'/q'$ are reduced fractions with positive denominators $q, q' < 2Q$, then

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{qq'} > \frac{1}{4Q^2}.$$

Consequently, an interval of length ℓ contains at most $4Q^2\ell + 1$ such fractions. Applying this to each I_ν and summing proves (2.1). $\square$

The complementary estimate supplies a positive proportion of reduced source fractions in every fixed interval.

Lemma 2.2 (Reduced fractions in a denominator block). *There exists an absolute constant $c_F > 0$ with the following property. For every nondegenerate compact interval $J \subseteq \mathbb{R}$, there exists $Q_0(J) \in \mathbb{Z}_{>0}$ such that*

$$(2.2) \quad \# \left\{ \frac{p}{q} \in J : \gcd(p, q) = 1, Q \leq q < 2Q \right\} \geq c_F |J| Q^2$$

for every integer $Q \geq Q_0(J)$. One may take $c_F = 1/4$.

Proof. Write $J = [\alpha, \beta]$ and, for $q \geq 1$, let

$$N_q(J) := \#\{p \in \mathbb{Z} : \alpha q \leq p \leq \beta q, \gcd(p, q) = 1\}.$$

By Möbius inversion,

$$N_q(J) = |J|\varphi(q) + O(\tau_0(q)),$$

where $\tau_0(q)$ denotes the number of positive divisors of q , and the implied constant is absolute. Hence, using the classical estimates [2],

$$\sum_{n \leq X} \varphi(n) = \frac{3}{\pi^2} X^2 + O(X \log X), \quad \sum_{n \leq X} \tau_0(n) = O(X \log X),$$

we obtain

$$\sum_{Q \leq q < 2Q} N_q(J) = \frac{9}{\pi^2} |J| Q^2 + O((1 + |J|) Q \log Q).$$

Since J is fixed and nondegenerate, the error term is $o(|J|Q^2)$. As $9/\pi^2 > 1/4$, the result follows for all sufficiently large Q . $\square$

3. WRONSKIAN LOCALIZATION AND UNIFORM SUBLEVEL BOUNDS

In Section 5, the determinant argument produces auxiliary polynomials $A(X) + B(X)Y$ whose coefficients depend on the rational points being counted. We therefore need sublevel estimates for $A(x) + B(x)f(x)$ that are uniform over all normalized coefficient vectors. Nonrationality of f provides the required Wronskian nondegeneracy after localization.

Let $\phi_1, \dots, \phi_N$ be functions of class C^{N-1} on an interval. Their *Wronskian* is

$$(3.1) \quad W(\phi_1, \dots, \phi_N)(x) := \det(\phi_j^{(k-1)}(x))_{1 \leq k, j \leq N}.$$

3.1. Rationality and linear independence. We first recall the classical Wronskian criterion for analytic functions. The analyticity assumption is essential here: the corresponding statement fails, in general, for smooth functions.

Lemma 3.1 (Analytic independence and Wronskians). *Let $U \subseteq \mathbb{R}$ be a connected open interval. If $\phi_1, \dots, \phi_N$ are real-analytic and linearly independent on U , then*

$$W(\phi_1, \dots, \phi_N) \not\equiv 0 \quad \text{on } U.$$

This is the classical analytic Wronskian criterion, going back at least to Bôcher [6]; see also [8] for a modern treatment.

For the remainder of the local argument, fix a real-analytic function $f : U \rightarrow \mathbb{R}$ which is not the restriction to U of a rational function in $\mathbb{R}(x)$. For $d \geq 0$, put

$$(3.2) \quad N_d := 2(d+1)$$

and consider the indexed family

$$(3.3) \quad \mathcal{F}_d(f) := (x^i f(x)^j)_{\substack{0 \leq i \leq d \\ j \in \{0,1\}}}.$$

Lemma 3.2 (Rationality and linear dependence). *Let $f : U \rightarrow \mathbb{R}$ be real-analytic. The following are equivalent.*

- (i) *f is the restriction to U of a rational function in $\mathbb{R}(x)$ with no pole in U ;*
- (ii) *the family $\mathcal{F}_d(f)$ is linearly dependent over $\mathbb{R}$ for some $d \geq 0$.*

Consequently, under the standing hypothesis on f , the family $\mathcal{F}_d(f)$ is linearly independent for every $d \geq 0$.

Proof. Suppose first that $f = P/Q$ on U , where $P, Q \in \mathbb{R}[X]$, Q has no zero in U , and $Q \neq 0$. If $d \geq \max\{\deg P, \deg Q\}$, then

$$Q(x)f(x) - P(x) = 0 \quad (x \in U),$$

which is a nontrivial linear relation among the members of $\mathcal{F}_d(f)$.

Conversely, suppose that $\mathcal{F}_d(f)$ is linearly dependent. Then there are polynomials $A, B \in \mathbb{R}[X]$, not both zero and of degree at most d , such that

$$(3.4) \quad A(x) + B(x)f(x) = 0 \quad (x \in U).$$

If $B = 0$, then A vanishes on the open interval U , and hence $A = 0$, a contradiction. Thus $B \neq 0$.

Let $D = \gcd(A, B)$ in $\mathbb{R}[X]$ and write $A = DA_0$, $B = DB_0$, with A_0 and B_0 coprime. On the nonempty open set $U \setminus \{D = 0\}$, equation (3.4) gives $A_0 + B_0 f = 0$. Since D has only finitely many zeros, continuity extends this identity to all of U :

$$A_0(x) + B_0(x)f(x) = 0 \quad (x \in U).$$

If $B_0(x_0) = 0$ at some $x_0 \in U$, then the last identity also gives $A_0(x_0) = 0$, contrary to the coprimality of A_0 and B_0 . Therefore B_0 has no zero in U , and

$$f(x) = -\frac{A_0(x)}{B_0(x)} \quad (x \in U).$$

This proves (i). □

Order the pairs

$$(i, j) \in \{0, \dots, d\} \times \{0, 1\}$$

lexicographically, and denote the corresponding ordered family of functions $x^i f(x)^j$ by

$$\varphi_{d,1}, \dots, \varphi_{d,N_d}.$$

Define

$$(3.5) \quad W_d(x) := W(\varphi_{d,1}, \dots, \varphi_{d,N_d})(x).$$

The ordering affects W_d at most by a sign and therefore has no bearing on its zero set.

Corollary 3.3. *For every $d \geq 0$, the function W_d is real-analytic and not identically zero on U . Consequently, if $I \subseteq U$ is a compact interval, then*

$$\{x \in I : W_d(x) = 0\}$$

is finite.

Proof. The functions in $\mathcal{F}_d(f)$ are real-analytic and, by Lemma 3.2, linearly independent. Lemma 3.1 therefore gives $W_d \not\equiv 0$ on U . The function W_d is real-analytic because it is a determinant of derivatives of real-analytic functions. If it had infinitely many zeros in I , compactness would give an accumulation point in U , and the identity theorem would force $W_d \equiv 0$ on the connected interval U , a contradiction. $\square$

3.2. One-dimensional sublevel estimates. We next record the elementary one-dimensional estimate underlying the uniform argument. It expresses the principle that a function with a derivative uniformly separated from zero cannot remain small on a long set.

Lemma 3.4. *Let $k \geq 1$, let $I \subseteq \mathbb{R}$ be a nondegenerate compact interval, and let $g \in C^k(I)$. Suppose that*

$$(3.6) \quad |g^{(k)}(x)| \geq \lambda > 0 \quad (x \in I).$$

Then, for every $\varepsilon > 0$, the set

$$E_g(\varepsilon) := \{x \in I : |g(x)| \leq \varepsilon\}$$

has at most $2k + 1$ connected components and satisfies

$$(3.7) \quad |E_g(\varepsilon)| \leq 2k(2k + 1) \left(\frac{\varepsilon}{\lambda}\right)^{1/k}.$$

Proof. Since $g^{(k)}$ is continuous and nowhere zero on I , it has constant sign. By Rolle's theorem, for every $c \in \mathbb{R}$ the equation $g(x) = c$ has at most k solutions. Hence the relative boundary of

$$E_g(\varepsilon) = \{x \in I : |g(x)| \leq \varepsilon\}$$

has at most $2k$ points, and $E_g(\varepsilon)$ has at most $2k + 1$ connected components.

Let $[a, b]$ be a nondegenerate component of $E_g(\varepsilon)$, and put

$$h := \frac{b - a}{k}, \quad x_j := a + jh \quad (0 \leq j \leq k).$$

By the mean-value theorem for finite differences, for some $\theta \in (a, b)$,

$$h^k g^{(k)}(\theta) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} g(x_j).$$

Since $|g(x_j)| \leq \varepsilon$,

$$h^k \lambda \leq 2^k \varepsilon,$$

and therefore

$$b - a = kh \leq 2k \left(\frac{\varepsilon}{\lambda} \right)^{1/k}.$$

Summing over at most $2k + 1$ nondegenerate components gives

$$|E_g(\varepsilon)| \leq 2k(2k + 1) \left(\frac{\varepsilon}{\lambda} \right)^{1/k}.$$

□

We now pass from a lower bound for one prescribed derivative to a bound that is uniform over all normalized linear combinations of a fixed finite family. Wronskian nondegeneracy gives a uniform lower bound for at least one derivative at each point. Compactness of the product of the coefficient sphere and the source interval makes the resulting local estimates uniform.

Lemma 3.5 (Uniform sublevel bounds under Wronskian nondegeneracy). *Let $J \subseteq \mathbb{R}$ be a nondegenerate compact interval, let $N \geq 2$, and let $\phi_1, \dots, \phi_N$ be of class C^{N-1} on a neighborhood of J . Assume that*

$$(3.8) \quad \inf_{x \in J} |W(\phi_1, \dots, \phi_N)(x)| > 0.$$

For $c = (c_1, \dots, c_N) \in \mathbb{R}^N$, write

$$G_c(x) = \sum_{j=1}^N c_j \phi_j(x).$$

Then there exist constants

$$C_{\text{sub}} > 0, \quad R_{\text{sub}} \in \mathbb{Z}_{>0}, \quad \varepsilon_0 \in (0, 1)$$

such that, for every $c \in \mathbb{R}^N$ with $\|c\|_2 = 1$ and every $0 < \varepsilon < \varepsilon_0$, the set

$$\{x \in J : |G_c(x)| \leq \varepsilon\}$$

is a union of at most R_{sub} intervals and satisfies

$$(3.9) \quad |\{x \in J : |G_c(x)| \leq \varepsilon\}| \leq C_{\text{sub}} \varepsilon^{1/(N-1)}.$$

Proof. Consider the jet matrix

$$\mathcal{A}(x) = (\phi_j^{(k)}(x))_{\substack{0 \leq k \leq N-1 \\ 1 \leq j \leq N}}.$$

Its determinant is the Wronskian in (3.8); hence $\mathcal{A}(x)$ is invertible for every $x \in J$. The least singular value of $\mathcal{A}(x)$ depends continuously on x . Since J is compact, there exists $\sigma > 0$ such that

$$(3.10) \quad \|\mathcal{A}(x)c\|_2 \geq \sigma \|c\|_2 \quad (x \in J, c \in \mathbb{R}^N).$$

For $\|c\|_2 = 1$,

$$\mathcal{A}(x)c = (G_c(x), G'_c(x), \dots, G_c^{(N-1)}(x))^T,$$

and therefore

$$\max_{0 \leq k \leq N-1} |G_c^{(k)}(x)| \geq \eta, \quad \eta := \frac{\sigma}{\sqrt{N}},$$

for every $(c, x) \in \mathbb{S}^{N-1} \times J$.

Fix $(c_0, x_0) \in \mathbb{S}^{N-1} \times J$. There is an index $k(c_0, x_0) \in \{0, \dots, N-1\}$ for which

$$\left| G_{c_0}^{(k(c_0, x_0))}(x_0) \right| \geq \eta.$$

Continuity in (c, x) yields a relatively open neighborhood U of c_0 in $\mathbb{S}^{N-1}$ and a compact interval $I \subseteq J$, with $x_0 \in \text{int}_J(I)$, such that

$$\left| G_c^{(k(c_0, x_0))}(x) \right| \geq \frac{\eta}{2} \quad (c \in U, x \in I).$$

These sets $U \times \text{int}_J(I)$ cover the compact space $\mathbb{S}^{N-1} \times J$. Choose a finite subcover

$$U_\nu \times \text{int}_J(I_\nu), \quad 1 \leq \nu \leq L,$$

and let $k_\nu \in \{0, \dots, N-1\}$ denote the corresponding derivative order. Thus

$$(3.11) \quad \left| G_c^{(k_\nu)}(x) \right| \geq \frac{\eta}{2} \quad (c \in U_\nu, x \in I_\nu).$$

The finite cover, and hence L and all constants derived from it, depend only on J and the family $\phi_1, \dots, \phi_N$, and not on the subsequent choice of $c \in \mathbb{S}^{N-1}$.

Fix $c \in \mathbb{S}^{N-1}$ and let

$$\mathcal{N}(c) := \{\nu \in \{1, \dots, L\} : c \in U_\nu\}.$$

Then the intervals I_ν , $\nu \in \mathcal{N}(c)$, cover J by their relative interiors. Put

$$\varepsilon_0 := \min \left\{ \frac{1}{2}, \frac{\eta}{4} \right\}.$$

If $0 < \varepsilon < \varepsilon_0$ and $k_\nu = 0$, then (3.11) shows that the sublevel set does not meet I_ν . If $1 \leq k_\nu \leq N-1$, Lemma 3.4, with $\lambda = \eta/2$, gives

$$|\{x \in I_\nu : |G_c(x)| \leq \varepsilon\}| \leq 2k_\nu(2k_\nu + 1) \left(\frac{2\varepsilon}{\eta} \right)^{1/k_\nu}.$$

Since $0 < \varepsilon < 1$ and $k_\nu \leq N-1$, this is at most

$$C_* \varepsilon^{1/(N-1)}, \quad C_* := \max_{1 \leq k \leq N-1} 2k(2k+1) \left(\frac{2}{\eta} \right)^{1/k}.$$

The same lemma shows that each such local sublevel set has at most $2N-1$ connected components.

Since $\#\mathcal{N}(c) \leq L$, subadditivity of length now gives

$$|\{x \in J : |G_c(x)| \leq \varepsilon\}| \leq LC_* \varepsilon^{1/(N-1)}.$$

Moreover, this set has at most $L(2N-1)$ connected components. Thus one may take

$$C_{\text{sub}} := LC_*, \quad R_{\text{sub}} := L(2N-1),$$

and these constants, as well as ε_0 , depend only on J and the family $\phi_1, \dots, \phi_N$, and not on c . $\square$

We specialize this result to the target-linear polynomials that arise from the determinant argument. For

$$P(X, Y) = A(X) + B(X)Y = \sum_{i=0}^d c_{i0}X^i + \sum_{i=0}^d c_{i1}X^iY,$$

write

$$\|P\|_2 := \left(\sum_{i=0}^d (c_{i0}^2 + c_{i1}^2) \right)^{1/2}.$$

Corollary 3.6 (Uniform rational-relation sublevel estimate). *Fix $d \geq 0$ and a nondegenerate compact interval $J \Subset U$ on which W_d does not vanish. Put $N = N_d = 2(d+1)$. Then there exist constants*

$$C_{d,J} > 0, \quad R_{d,J} \in \mathbb{Z}_{>0}, \quad \varepsilon_{d,J} > 0$$

such that, for every polynomial

$$P(X, Y) = A(X) + B(X)Y, \quad \deg A, \deg B \leq d, \quad \|P\|_2 = 1,$$

and every $0 < \varepsilon < \varepsilon_{d,J}$, the set

$$\{x \in J : |P(x, f(x))| \leq \varepsilon\}$$

is a union of at most $R_{d,J}$ intervals and satisfies

$$(3.12) \quad |\{x \in J : |P(x, f(x))| \leq \varepsilon\}| \leq C_{d,J} \varepsilon^{1/(N-1)}.$$

The constants depend only on d , J , and the restriction of f to a neighborhood of J .

Proof. Arrange the coefficients (c_{i0}, c_{i1}) in the same order as the family used to define W_d . Then $P(x, f(x))$ is the corresponding linear combination of the members of $\mathcal{F}_d(f)$, and its coefficient vector has Euclidean norm one. Since W_d is continuous, nonzero on the compact interval J , and hence bounded away from zero, Lemma 3.5 applies and gives the assertion. $\square$

4. ANISOTROPIC DETERMINANT ESTIMATES

We compare determinants formed from rational points close to the graph of f . Smoothness bounds these determinants from above on short source intervals, while clearing the source and target denominators separately gives a lower bound for every nonzero determinant. Their comparison will force an auxiliary relation of degree one in the target variable.

For vectors $v_1, \dots, v_s \in \mathbb{R}^N$, we equip the exterior power $\bigwedge^s \mathbb{R}^N$ with its Euclidean norm. Thus

$$\|v_1 \wedge \dots \wedge v_s\|^2 = \det(\langle v_i, v_j \rangle)_{1 \leq i, j \leq s}.$$

In particular, when $s = N$,

$$\|v_1 \wedge \dots \wedge v_N\| = |\det(v_1, \dots, v_N)|.$$

We shall also use

$$\|v \wedge \omega\| \leq \|v\| \|\omega\|$$

for $v \in \mathbb{R}^N$ and $\omega \in \bigwedge^s \mathbb{R}^N$.

4.1. Exterior decay along a smooth curve. Several values of a smooth one-dimensional curve taken in an interval of length ρ become increasingly dependent as $\rho \rightarrow 0$. The exponent below reflects the successive Taylor orders $0, 1, \dots, s-1$.

Lemma 4.1. *Let $J \subseteq \mathbb{R}$ be a nondegenerate compact interval, let $1 \leq s \leq N$, and let*

$$\Phi : J \longrightarrow \mathbb{R}^N$$

be the restriction to J of a C^s map defined on a neighborhood of J . There exists a constant $C_\Phi > 0$ such that, whenever $0 < \rho \leq 1$ and $t_1, \dots, t_s$ lie in a common subinterval of J of length at most ρ , one has

$$(4.1) \quad \|\Phi(t_1) \wedge \dots \wedge \Phi(t_s)\| \leq C_\Phi \rho^{s(s-1)/2}.$$

Proof. Let

$$a := \min_{1 \leq i \leq s} t_i, \quad \ell := \max_{1 \leq i \leq s} t_i - a.$$

If $\ell = 0$, the assertion is immediate: for $s \geq 2$ the exterior product vanishes, while for $s = 1$ it follows after taking $C_\Phi \geq \sup_J \|\Phi\|$. Assume henceforth that $0 < \ell \leq \rho$. Taylor's formula with integral remainder gives

$$(4.2) \quad \Phi(t_i) = \sum_{r=0}^{s-1} (t_i - a)^r v_r + R_i, \quad v_r := \frac{\Phi^{(r)}(a)}{r!},$$

where

$$R_i = \frac{1}{(s-1)!} \int_a^{t_i} (t_i - u)^{s-1} \Phi^{(s)}(u) du.$$

Since $\Phi^{(s)}$ is bounded on a neighborhood of J , there is a constant $C_0 > 0$, independent of a , ℓ , and i , such that

$$(4.3) \quad \|R_i\| \leq C_0 |t_i - a|^s \leq C_0 \ell^s.$$

Also $|t_i - a|^r \leq \ell^r$ for $0 \leq r \leq s-1$.

Expand the exterior product of the expressions in (4.2). Consider a term containing exactly k remainder vectors. The remaining $s-k$ factors are chosen from $v_0, \dots, v_{s-1}$. A term in which some v_r occurs twice vanishes. Thus, in every nonzero term, the corresponding indices $r_1, \dots, r_{s-k}$ are distinct and

$$r_1 + \dots + r_{s-k} \geq 0 + 1 + \dots + (s-k-1) = \frac{(s-k)(s-k-1)}{2}.$$

Such a term is therefore bounded by a constant times

$$\ell^{ks + (s-k)(s-k-1)/2}.$$

The identity

$$ks + \frac{(s-k)(s-k-1)}{2} = \frac{s(s-1)}{2} + \frac{k(k+1)}{2}$$

shows that this is $O_\Phi(\ell^{s(s-1)/2})$. There are only finitely many terms, and the vectors $v_0, \dots, v_{s-1}$ are uniformly bounded as a ranges over J . Hence

$$\|\Phi(t_1) \wedge \dots \wedge \Phi(t_s)\| \leq C_\Phi \ell^{s(s-1)/2} \leq C_\Phi \rho^{s(s-1)/2}.$$

□

4.2. The perturbed determinant. For $d \geq 0$, put $N = 2(d+1)$, order the pairs

$$(i, j) \in \{0, \dots, d\} \times \{0, 1\}$$

as in Section 3, and define

$$(4.4) \quad \Psi_d(x, y) := (x^i y^j)_{\substack{0 \leq i \leq d \\ j \in \{0, 1\}}} \in \mathbb{R}^N, \quad \Phi_d(x) := \Psi_d(x, f(x)).$$

Lemma 4.2 (Perturbed target-linear determinant). *Let $d \geq 0$, put $N = 2(d+1)$, and let $J \Subset U$ be a nondegenerate compact interval. There exists a constant $C_{d,J}^{\text{pert}} > 0$, depending only on d , J , and the restriction of f to a neighborhood of J , with the following property. Suppose that $0 < \rho \leq 1$, that $x_1, \dots, x_N \in J$ lie in a common subinterval of J of length at most ρ , and that $y_1, \dots, y_N \in \mathbb{R}$ satisfy*

$$(4.5) \quad |y_k - f(x_k)| \leq \delta \leq \rho^N \quad (1 \leq k \leq N).$$

Regarding the vectors $\Psi_d(x_k, y_k)$ as the rows of an $N \times N$ matrix, one has

$$(4.6) \quad \left| \det(\Psi_d(x_k, y_k))_{1 \leq k \leq N} \right| \leq C_{d,J}^{\text{pert}} \rho^{N(N-1)/2}.$$

Proof. Set

$$E_k := \Psi_d(x_k, y_k) - \Psi_d(x_k, f(x_k)).$$

The coordinates of E_k corresponding to $j = 0$ vanish, whereas those corresponding to $j = 1$ are

$$x_k^i (y_k - f(x_k)), \quad 0 \leq i \leq d.$$

Since $x_k \in J$, there is a constant $C_0 = C_0(d, J)$ such that

$$(4.7) \quad \|E_k\| \leq C_0 \delta \quad (1 \leq k \leq N).$$

We have

$$\Psi_d(x_k, y_k) = \Phi_d(x_k) + E_k.$$

Expanding the exterior product of these N vectors by multilinearity, consider a term containing exactly r error vectors. If $r < N$, the exterior norm inequality, (4.7), and Lemma 4.1, applied to the remaining $N - r$ curve vectors, bound this term by

$$C_r \delta^r \rho^{(N-r)(N-r-1)/2}.$$

For $r = N$, the same bound holds with the standard convention $\bigwedge^0 \mathbb{R}^N = \mathbb{R}$ and $\|1\| = 1$. Absorbing the finitely many combinatorial factors, we obtain

$$(4.8) \quad \left| \det(\Psi_d(x_k, y_k))_{1 \leq k \leq N} \right| \leq C \sum_{r=0}^N \delta^r \rho^{(N-r)(N-r-1)/2}.$$

By (4.5), the summand of index r is at most

$$\rho^{rN+(N-r)(N-r-1)/2}.$$

The exact identity

$$(4.9) \quad rN + \frac{(N-r)(N-r-1)}{2} = \frac{N(N-1)}{2} + \frac{r(r+1)}{2}$$

shows that, since $0 < \rho \leq 1$, every summand is at most $\rho^{N(N-1)/2}$. Summing over $0 \leq r \leq N$ proves (4.6). $\square$

4.3. The arithmetic lower bound. The preceding estimate is purely analytic. Rationality supplies the complementary fact that a nonzero target-linear determinant cannot be too small.

Lemma 4.3. *Fix $d \geq 0$ and put $N = 2(d+1)$. Let*

$$x_k = \frac{p_k}{q_k}, \quad y_k = \frac{a_k}{b_k}, \quad 1 \leq k \leq N,$$

where $p_k, a_k \in \mathbb{Z}$ and $q_k, b_k \in \mathbb{Z}_{>0}$. The fractions need not be reduced. Define

$$\Delta := \det(x_k^i y_k^j)_{\substack{1 \leq k \leq N \\ 0 \leq i \leq d, j \in \{0,1\}}}.$$

If $\Delta \neq 0$, then

$$(4.10) \quad |\Delta| \geq \prod_{k=1}^N q_k^{-d} b_k^{-1}.$$

In particular, if $q_k < 2Q$ and $b_k < 2B$ for every k , then

$$(4.11) \quad |\Delta| \geq (2Q)^{-dN} (2B)^{-N}.$$

Proof. Multiply the k th row of the matrix defining Δ by $q_k^d b_k$. The entry corresponding to (i, j) becomes

$$q_k^d b_k x_k^i y_k^j = \begin{cases} p_k^i q_k^{d-i} b_k, & j = 0, \\ p_k^i q_k^{d-i} a_k, & j = 1, \end{cases}$$

and is therefore an integer. The determinant of the resulting integer matrix is

$$\left(\prod_{k=1}^N q_k^d b_k \right) \Delta.$$

If $\Delta \neq 0$, this is a nonzero integer and hence has absolute value at least one, proving (4.10). The final assertion follows from $q_k < 2Q$ and $b_k < 2B$. $\square$

The next section combines the analytic upper bound (4.6) with the arithmetic lower bound (4.11). The two estimates retain the source and target heights separately; their balance is achieved by adapting the source degree to the relative size of these two heights.

5. TWO-HEIGHT COUNTING ON ANALYTIC GRAPHS

We now establish the main counting estimate for the nonrational real-analytic function $f : U \rightarrow \mathbb{R}$ fixed in Section 3. We count rational source points p/q , with $q \asymp Q$, whose images admit exceptionally good rational approximations, keeping the source and target denominator scales separate. The numerical parameters below are chosen to ensure the strict exponent inequalities needed in the proof; we make no attempt to optimize them.

We fix

$$(5.1) \quad \gamma = 10, \quad \tau = 100, \quad v = \frac{2}{\gamma} = \frac{1}{5}, \quad \tau - 3 = 97.$$

The target degree is one throughout this section. The parameter γ controls the source degree, τ is the approximation exponent that will eventually be excluded for the image, and v separates the elementary and determinant ranges of target denominators.

For every integer $A \geq 3$, define

$$(5.2) \quad D(A) := \max \left\{ 2, \left\lceil \frac{\gamma A}{\tau - 3} \right\rceil \right\} = \max \left\{ 2, \left\lceil \frac{10A}{97} \right\rceil \right\},$$

and, for $Q \geq 2$, put

$$(5.3) \quad T(Q, A) := Q^{A/(\tau-3)} = Q^{A/97}.$$

The quantity A measures the required accuracy in the source variable, whereas $T(Q, A)$ is the largest target height treated at source scale Q . The exponent $\tau - 3$ is chosen for the subsequent fusion: for fixed A , $T(Q, A)^{2-\tau} = o(Q^{-A})$ as $Q \rightarrow \infty$. Thus the contribution from the next lower target cutoff can be made small relative to the next source interval.

Let $I_* \Subset U$ be a nondegenerate compact interval on which

$$(5.4) \quad 0 < m_* \leq |f'(x)| \leq M_* \quad (x \in I_*).$$

Since f' is continuous and does not vanish on I_* , it has constant sign there. Hence $f|_{I_*}$ is strictly monotone, and the mean-value theorem gives

$$(5.5) \quad |x_2 - x_1| \leq m_*^{-1} |f(x_2) - f(x_1)| \quad (x_1, x_2 \in I_*).$$

For a nondegenerate compact interval $J \subseteq I_*$, integers $A \geq 3$ and $H \geq 2$, and an integer $Q \geq 2$, define the set of *resonant source centers*

$$(5.6) \quad \begin{aligned} \mathcal{D}(J; Q, A, H) := & \left\{ \frac{p}{q} \in J : \gcd(p, q) = 1, \quad Q \leq q < 2Q, \right. \\ & \exists a \in \mathbb{Z}, \quad b \in \mathbb{Z}_{>0} \text{ with } H \leq b < T(Q, A), \\ & \left. |f(p/q) - a/b| \leq 2b^{-\tau} + 4M_*Q^{-A} \right\}. \end{aligned}$$

The source fractions are reduced, whereas the target fractions need not be. For $p/q \in \mathcal{D}(J; Q, A, H)$, any pair (a, b) satisfying the conditions in (5.6) will be called a *witness* for p/q . The term Q^{-A} is a stability margin which will allow avoidance at a rational center to persist on the next source interval.

Proposition 5.1 (Two-height counting on analytic graphs). *There exists a constant $C_{\text{low}} > 0$, depending only on f , I_* , m_* , M_* and the fixed parameters in (5.1), with the following property. Let $J \subseteq I_*$ be a nondegenerate compact interval and let $A \geq 3$ be an integer. Assume that*

$$(5.7) \quad W_d(x) \neq 0 \quad (x \in J, 2 \leq d \leq D(A)).$$

Then there exist constants

$$C(J, A) > 0 \quad \text{and} \quad Q_0(J, A) \in \mathbb{Z}_{>0}$$

such that, for every integer $H \geq 2$ and every integer $Q \geq Q_0(J, A)$,

$$(5.8) \quad \#\mathcal{D}(J; Q, A, H) \leq C_{\text{low}} Q^2 H^{-98} + C(J, A) Q^{17/10}.$$

In particular, the constants $C(J, A)$ and $Q_0(J, A)$ are uniform in H .

Proof. Fix J and A as in the statement. Every lower bound imposed on Q below will depend at most on J and A , and will be independent of H and of the target block.

If $T(Q, A) \leq H$, then $\mathcal{D}(J; Q, A, H) = \emptyset$. Assume henceforth that $T(Q, A) > H$. For every integer $k \geq 0$ such that

$$B_k := 2^k H < T(Q, A),$$

put

$$(5.9) \quad \mathcal{B}_k := [B_k, \min\{2B_k, T(Q, A)\}) \cap \mathbb{Z}_{>0}.$$

These sets form a disjoint partition of $[H, T(Q, A)) \cap \mathbb{Z}_{>0}$. Let $\mathcal{D}_k$ be the set of source centers admitting at least one witness (a, b) with $b \in \mathcal{B}_k$. Then

$$(5.10) \quad \mathcal{D}(J; Q, A, H) \subseteq \bigcup_k \mathcal{D}_k, \quad \#\mathcal{D}(J; Q, A, H) \leq \sum_k \#\mathcal{D}_k.$$

A center may occur in more than one $\mathcal{D}_k$, which only produces an admissible overcount.

A block with left endpoint $B = B_k$ is called *small* if $B < Q^v$ and *large* if $B \geq Q^v$. A block beginning below Q^v and crossing that threshold is assigned to the small range; the estimates there use only $B \leq b < 2B$.

Small target heights. Set

$$(5.11) \quad F_* := \max_{x \in I_*} |f(x)|, \quad C_* := F_* + 2^{1-\tau} + \frac{M_*}{2}.$$

Suppose that $x \in I_*$, $b \geq 2$, $A \geq 3$, $Q \geq 2$, and

$$|f(x) - a/b| \leq 2b^{-\tau} + 4M_*Q^{-A}.$$

Since $Q^{-A} \leq 2^{-3}$,

$$\frac{|a|}{b} \leq F_* + 2b^{-\tau} + 4M_*Q^{-A} \leq C_*.$$

Thus $|a| \leq C_*b$. For $B \leq b < 2B$, there are therefore $O(B^2)$ pairs (a, b) that can occur as witnesses, with an implied constant depending only on C_* .

For each such pair, let

$$E_{a,b} := \left\{ x \in J : |f(x) - a/b| \leq 2b^{-\tau} + 4M_*Q^{-A} \right\}.$$

Because f is strictly monotone on I_* , the set $E_{a,b}$ is empty, a singleton, or an interval. If $x_1 < x_2$ belong to $E_{a,b}$, then (5.5) gives

$$m_*(x_2 - x_1) \leq |f(x_2) - f(x_1)| \leq 2(2b^{-\tau} + 4M_*Q^{-A}).$$

Consequently,

$$(5.12) \quad |E_{a,b}| \leq \frac{2}{m_*}(2b^{-\tau} + 4M_*Q^{-A}).$$

Let E_B be the union of the sets $E_{a,b}$ over one small target block. It is a union of $O(B^2)$ intervals, and

$$(5.13) \quad |E_B| \ll B^{2-\tau} + B^2Q^{-A}.$$

Every center in the corresponding $\mathcal{D}_k$ belongs to E_B . Applying Lemma 2.1 gives

$$(5.14) \quad \begin{aligned} \#\mathcal{D}_k &\ll Q^2|E_B| + B^2 \\ &\ll Q^2B^{2-\tau} + Q^{2-A}B^2 + B^2. \end{aligned}$$

We sum over the dyadic left endpoints $B = 2^kH < Q^v$. Since $2 - \tau < 0$,

$$\sum_{2^kH < Q^v} (2^kH)^{2-\tau} \ll H^{2-\tau},$$

whereas

$$\sum_{2^kH < Q^v} (2^kH)^2 \ll Q^{2v}.$$

It follows that

$$(5.15) \quad \sum_{B_k < Q^v} \#\mathcal{D}_k \leq C_{\text{low}}Q^2H^{2-\tau} + O(Q^{2-A+2v} + Q^{2v}).$$

The coefficient C_{low} depends only on f , I_* , m_* , M_* and the fixed parameters in (5.1). Since $A \geq 3$ and $v = 1/5$,

$$2 - A + 2v \leq -\frac{3}{5}, \quad 2v = \frac{2}{5}.$$

Thus

$$(5.16) \quad \sum_{B_k < Q^v} \#\mathcal{D}_k \leq C_{\text{low}}Q^2H^{2-\tau} + O(Q^{2/5}).$$

Large target heights. If there are no blocks with $B_k \geq Q^v$, there is nothing to prove. This is automatically the case when $A \leq 19$, since then

$$T(Q, A) = Q^{A/97} < Q^{1/5} = Q^v.$$

Hence, in the large-target range one necessarily has $A \geq 20$. Fix one such block, write $B = B_k$, and define

$$(5.17) \quad u := \frac{\log B}{\log Q}, \quad B = Q^u.$$

Since $Q^v \leq B < T(Q, A)$,

$$(5.18) \quad \frac{1}{5} = v \leq u < \frac{A}{97}.$$

We shall show that each large target block contributes $O_{J,A}(Q^{33/20})$ resonant centers: we partition J into $O_J(Q^\kappa)$ short source cells and prove that each cell contains only $O_{J,A}(1)$ resonant centers. Set

$$(5.19) \quad d := \lceil \gamma u \rceil = \lceil 10u \rceil, \quad N := 2(d+1), \quad \kappa := \frac{3(d+u)}{N-1},$$

and put

$$(5.20) \quad \rho := Q^{-\kappa}.$$

The degree $d \asymp u$ adapts the source degree to the relative target height $B = Q^u$, while the choice of κ gives

$$\rho^{N(N-1)/2} = Q^{-\frac{3}{2}N(d+u)},$$

to be compared below with the arithmetic scale $Q^{-N(d+u)}$. Because $10u \geq 2$, we have $d \geq 2$. Moreover, $10u < 10A/97$, and hence $d \leq D(A)$. Thus (5.7) applies to the chosen degree.

Partition J into half-open cells of length at most ρ , taking the final cell to be closed at its right endpoint. Every point of J then belongs to exactly one cell, and the number of cells is at most

$$(5.21) \quad 1 + |J|\rho^{-1} = 1 + |J|Q^\kappa \leq (1 + |J|)Q^\kappa.$$

Fix a cell C . For every center $x = p/q \in \mathcal{D}_k \cap C$, choose once and for all a witness (a, b) with $b \in \mathcal{B}_k$, and put $y = a/b$. Define

$$(5.22) \quad s := \min\{u\tau, A\}.$$

Since $b \geq B = Q^u$,

$$(5.23) \quad \begin{aligned} |y - f(x)| &\leq 2B^{-\tau} + 4M_*Q^{-A} \\ &\leq K_0Q^{-s}, \quad K_0 := 2 + 4M_*. \end{aligned}$$

We record uniform exponent bounds. From $d = \lceil 10u \rceil$ and $u \geq 1/5$,

$$(5.24) \quad 10u \leq d \leq 10u + 1 \leq 15u, \quad d + u \leq 16u.$$

Since $u \leq d/10$ and $N - 1 = 2d + 1$,

$$(5.25) \quad \kappa = \frac{3(d+u)}{2d+1} \leq \frac{3(11d/10)}{2d} = \frac{33}{20}.$$

The upper bound in (5.18) gives $97u < A$, while $\tau u = 100u$. Therefore

$$(5.26) \quad s \geq 97u.$$

Also, using $N/(N-1) \leq 2$ and (5.24),

$$(5.27) \quad \kappa N = 3(d+u) \frac{N}{N-1} \leq 6(d+u) \leq 96u.$$

Consequently,

$$(5.28) \quad s - \kappa N \geq u \geq \frac{1}{5}.$$

Finally,

$$(5.29) \quad N - 1 = 2d + 1 \leq 30u + 1 \leq 35u.$$

Thus the three numerical inequalities driving the large-target argument are

$$\kappa \leq \frac{33}{20}, \quad s - \kappa N \geq \frac{1}{5}, \quad \frac{s}{N-1} \geq \frac{97}{35} > 2.$$

They control, respectively, the number of source cells, the perturbation required in Lemma 4.2, and the Farey count within each cell.

Determinant comparison. For each integer d with $2 \leq d \leq D(A)$, let $C_{d,J}^{\text{pert}}$ be the constant in Lemma 4.2, and set

$$(5.30) \quad K_{\det}(J, A) := \max_{2 \leq d \leq D(A)} C_{d,J}^{\text{pert}} < \infty.$$

Since $D(A)$ is fixed once A is fixed, this maximum is taken over a finite set of constants determined before the choice of Q and of the target block. In particular, $K_{\det}(J, A)$ is independent of Q , u , B , and H .

By (5.23) and (5.28),

$$K_0 Q^{-s} \leq K_0 Q^{-\kappa N - 1/5}.$$

Hence $Q \geq K_0^5$ implies

$$K_0 Q^{-s} \leq Q^{-\kappa N} = \rho^N.$$

This restriction is simultaneous for all large target blocks and is independent of u , B , and H .

Suppose first that C contains at least N resonant centers. Choose N distinct centers and their fixed witnessing pairs, say (x_r, y_r) , $1 \leq r \leq N$, and define

$$\Delta := \det(x_r^i y_r^j)_{\substack{1 \leq r \leq N \\ 0 \leq i \leq d, j \in \{0,1\}}}.$$

All the x_r lie in the closure of a cell of length at most ρ . Lemma 4.2 and the definition of κ give

$$(5.31) \quad |\Delta| \leq K_{\det}(J, A) \rho^{N(N-1)/2} = K_{\det}(J, A) Q^{-\frac{3}{2}N(d+u)}.$$

If $\Delta \neq 0$, then $q_r < 2Q$ and $b_r < 2B$, so Lemma 4.3 gives

$$(5.32) \quad |\Delta| \geq (2Q)^{-dN} (2B)^{-N} = 2^{-N(d+1)} Q^{-N(d+u)}.$$

The two bounds are incompatible whenever

$$(5.33) \quad Q^{N(d+u)/2} > K_{\det}(J, A) 2^{N(d+1)}.$$

This threshold is uniform in the target block. Indeed, $d \geq 2$, $u \geq 1/5$, and $N \geq 6$, so

$$\frac{N(d+u)}{2} \geq \frac{6(2+1/5)}{2} = \frac{33}{5} > 6.$$

Since $N(d+1) = 2(d+1)^2$, it is enough to impose

$$Q^6 > K_{\det}(J, A) \max_{2 \leq d \leq D(A)} 2^{2(d+1)^2}.$$

This condition depends only on J and A . Thus, for all sufficiently large Q , the determinant associated with every choice of N distinct resonant centers in every cell is zero.

Let M_C be the number of resonant centers in C for the present target block. If $M_C < N$, then

$$M_C < N = 2(d+1) \leq 2(D(A)+1),$$

so $M_C = O_A(1)$. Assume that $M_C \geq N$ and enumerate the centers together with their chosen witnesses as

$$(x_r, y_r), \quad 1 \leq r \leq M_C,$$

where $x_r = p_r/q_r$ and $y_r = a_r/b_r$. Consider the $M_C \times N$ evaluation matrix

$$(5.34) \quad \mathcal{M}_C := (x_r^i y_r^j)_{\substack{1 \leq r \leq M_C \\ 0 \leq i \leq d, j \in \{0,1\}}}.$$

Since the preceding choice of N distinct centers was arbitrary, every $N \times N$ minor of $\mathcal{M}_C$ vanishes. Hence $\text{rank } \mathcal{M}_C \leq N-1$, so the right kernel of $\mathcal{M}_C$ in $\mathbb{R}^N$ is nontrivial. Choose a unit vector (c_{i0}, c_{i1}) in this right kernel and define

$$(5.35) \quad P_C(X, Y) := A_C(X) + B_C(X)Y := \sum_{i=0}^d c_{i0} X^i + \sum_{i=0}^d c_{i1} X^i Y, \quad \|P_C\|_2 = 1.$$

Then

$$(5.36) \quad P_C(x_r, y_r) = 0 \quad (1 \leq r \leq M_C).$$

Counting in one cell. Put

$$X_* := \max \left\{ 1, \max_{x \in J} |x| \right\}$$

and

$$(5.37) \quad L_{J,A} := \max_{2 \leq d \leq D(A)} \left(\sum_{i=0}^d X_*^{2i} \right)^{1/2} < \infty.$$

The coefficient normalization in (5.35) and Cauchy–Schwarz give $|B_C(x)| \leq L_{J,A}$ for $x \in J$. Since P_C is linear in Y , (5.36) and (5.23) imply

$$(5.38) \quad |P_C(x_r, f(x_r))| = |B_C(x_r)| |f(x_r) - y_r| \leq L_{J,A} K_0 Q^{-s}.$$

For $2 \leq d \leq D(A)$, let $C_{d,J}$, $R_{d,J}$, and $\varepsilon_{d,J}$ be supplied by Corollary 3.6, and set

$$(5.39) \quad \begin{aligned} C_{J,A}^{\text{sub}} &:= \max_{2 \leq d \leq D(A)} C_{d,J}, \\ R_{J,A}^{\text{sub}} &:= \max_{2 \leq d \leq D(A)} R_{d,J}, \\ \varepsilon_{J,A}^{\text{sub}} &:= \min_{2 \leq d \leq D(A)} \varepsilon_{d,J} > 0. \end{aligned}$$

These extrema are finite. Since $s \geq 97/5$,

$$L_{J,A} K_0 Q^{-s} \leq L_{J,A} K_0 Q^{-97/5},$$

so, after increasing the lower threshold for Q depending only on J and A , we have

$$(5.40) \quad L_{J,A} K_0 Q^{-s} < \varepsilon_{J,A}^{\text{sub}}.$$

This threshold is again uniform over all large target blocks. Define

$$S_C := \{x \in J : |P_C(x, f(x))| \leq L_{J,A} K_0 Q^{-s}\} \cap \overline{C}.$$

Intersecting the sublevel set from Corollary 3.6 with the interval $\overline{C}$ does not increase its number of connected components. Every resonant center assigned to C belongs to S_C . By Corollary 3.6,

$$(5.41) \quad |S_C| \ll_{J,A} Q^{-s/(N-1)},$$

and S_C is a union of at most $R_{J,A}^{\text{sub}}$ intervals. From (5.26) and (5.29),

$$(5.42) \quad \frac{s}{N-1} \geq \frac{97}{35} > 2.$$

The resonant centers are distinct reduced fractions with denominators in $[Q, 2Q]$. By (5.41) and (5.42),

$$Q^2 |S_C| \ll_{J,A} Q^{2-s/(N-1)} \leq Q^{2-97/35} = Q^{-27/35}.$$

Lemma 2.1 therefore gives

$$(5.43) \quad \begin{aligned} M_C &\leq 4Q^2 |S_C| + R_{J,A}^{\text{sub}} \\ &\ll_{J,A} Q^{-27/35} + 1 \ll_{J,A} 1. \end{aligned}$$

Together with the case $M_C < N$, this bound holds uniformly for every cell and every large target block.

Summation over cells and target blocks. By (5.21), there are $O_J(Q^\kappa)$ source cells, and (5.43) gives $O_{J,A}(1)$ resonant centers in each. Hence, by (5.25), one large target block contributes

$$(5.44) \quad O_{J,A}(Q^\kappa) = O_{J,A}(Q^{33/20}).$$

The total number of dyadic target blocks is at most

$$1 + \log_2 \frac{T(Q, A)}{H} \leq 1 + \frac{A}{97} \log_2 Q = O_A(\log Q).$$

Since $\log Q \ll Q^{1/20}$ for $Q \geq 2$, summing (5.44) gives

$$(5.45) \quad \sum_{B_k \geq Q^v} \#\mathcal{D}_k = O_{J,A}(Q^{33/20} \log Q) = O_{J,A}(Q^{17/10}).$$

Combining (5.10), (5.16), and (5.45), we obtain (5.8). We finally record the uniformity. For fixed J and A , only the finite range $2 \leq d \leq D(A)$ occurs, and the bounds $u \geq 1/5$ and $s - \kappa N \geq 1/5$ make all lower restrictions on Q in the large-height argument uniform in the target block. Thus none depends on H , u , B , the cell, or the witnessing pairs. Taking their maximum defines $Q_0(J, A)$, and enlarging the final implied constant defines $C(J, A)$. $\square$

6. FROM TWO-HEIGHT COUNTING TO RIGIDITY

6.1. A safe center. The counting estimate becomes effective once its quadratic main term is smaller than the supply of reduced source fractions. For the deduction below, the precise exponent $17/10$ is not essential: what matters is that the second term in Proposition 5.1 is $o(Q^2)$ as $Q \rightarrow \infty$, uniformly in the lower target cutoff H , while $H^{2-\tau} \rightarrow 0$ as $H \rightarrow \infty$. These two features produce safe rational centers at every sufficiently large source scale.

Lemma 6.1 (A safe center in one source block). *Assume the hypotheses and notation of Proposition 5.1. Let $H \geq 2$ be an integer and suppose in addition that*

$$(6.1) \quad C_{\text{low}} H^{2-\tau} \leq \frac{c_F}{4} |J|.$$

Then, for every sufficiently large integer Q , there exists a reduced fraction

$$r = \frac{p}{q} \in J, \quad Q \leq q < 2Q,$$

such that

$$(6.2) \quad |f(r) - a/b| > 2b^{-\tau} + 4M_* Q^{-A}$$

for every $a \in \mathbb{Z}$ and every integer b satisfying

$$H \leq b < T(Q, A).$$

Proof. By Lemma 2.2, for every sufficiently large Q , the interval J contains at least

$$c_F |J| Q^2$$

reduced fractions with denominator in $[Q, 2Q)$. On the other hand, Proposition 5.1 and (6.1) give

$$\#\mathcal{D}(J; Q, A, H) \leq \frac{c_F}{4} |J| Q^2 + C(J, A) Q^{17/10}.$$

Since $17/10 < 2$ and J, A are fixed, there exists $Q_1(J, A)$ such that, for every $Q \geq Q_1(J, A)$,

$$C(J, A)Q^{17/10} \leq \frac{c_F}{4}|J|Q^2.$$

Thus $\mathcal{D}(J; Q, A, H)$ does not exhaust the reduced fractions $p/q \in J$ with $Q \leq q < 2Q$. Hence any such fraction outside $\mathcal{D}(J; Q, A, H)$ satisfies (6.2) by definition. $\square$

6.2. Fusion and proof of Theorem 1.2. Fix a nonempty open interval $V \subseteq U$. Since f is not rational, it is nonconstant, so $f' \not\equiv 0$. As f' is real-analytic, its zeros are isolated. We may therefore choose a nondegenerate compact interval

$$(6.3) \quad I_* \Subset V$$

on which f' does not vanish. By continuity and compactness, there are constants $m_*, M_* > 0$ such that

$$0 < m_* \leq |f'(x)| \leq M_* \quad (x \in I_*),$$

as required in (5.4).

For $n \geq 0$, put

$$(6.4) \quad A_n := n + 3$$

and define

$$(6.5) \quad \mathcal{Z}_n := \bigcup_{2 \leq d \leq D(A_n)} \{x \in I_* : W_d(x) = 0\}.$$

Each $\mathcal{Z}_n$ is finite by Corollary 3.3. Since $D(A)$ is nondecreasing,

$$(6.6) \quad \mathcal{Z}_n \subseteq \mathcal{Z}_{n+1}.$$

Choose a nondegenerate compact interval

$$(6.7) \quad I_0 \subseteq \text{int}(I_*) \setminus \mathcal{Z}_0.$$

For a nondegenerate compact interval $I = [\alpha, \beta]$, let

$$(6.8) \quad J(I) := \left[\alpha + \frac{\beta - \alpha}{3}, \beta - \frac{\beta - \alpha}{3} \right]$$

be its middle third. Whenever I_n is defined, write $J_n := J(I_n)$. Then

$$(6.9) \quad |J_n| = \frac{1}{3}|I_n|, \quad \text{dist}(J_n, \mathbb{R} \setminus I_n) = \frac{1}{3}|I_n|.$$

Choose an integer $T_0 \geq 2$ such that

$$(6.10) \quad C_{\text{low}} T_0^{2-\tau} \leq \frac{c_F}{4}|J_0|.$$

This is possible because $2 - \tau = -98 < 0$.

We construct nondegenerate compact intervals I_n , strictly increasing positive integers T_n , and reduced fractions $r_n = p_n/q_n$ satisfying the following properties whenever the objects involved are defined:

$$(F1) \quad I_{n+1} \subset \text{int}(I_n);$$

- (F2) $I_n \cap \mathcal{Z}_n = \emptyset$;
- (F3) $T_{n+1} > T_n$;
- (F4) $q_n > 2q_{n-1}$ for $n \geq 1$;
- (F5) for every $x \in I_{n+1}$,

$$(6.11) \quad 0 < |x - r_n| < q_n^{-A_n};$$

- (F6) for every $x \in I_{n+1}$, every $a \in \mathbb{Z}$, and every integer b with $T_n \leq b < T_{n+1}$,

$$(6.12) \quad |f(x) - a/b| > b^{-\tau};$$

- (F7)

$$(6.13) \quad C_{\text{low}} T_n^{2-\tau} \leq \frac{c_F}{4} |J_n|.$$

Conditions (F5) and (F6) impose source approximation and target avoidance. Conditions (F2) and (F7) retain the hypotheses needed to choose the next safe center.

For clarity, we write $(F1)_n$, $(F3)_n$, $(F5)_n$, and $(F6)_n$ for the corresponding properties linking stage n to stage $n+1$, while $(F2)_n$ and $(F7)_n$ refer to the properties at stage n itself; similarly, $(F4)_n$ is understood for $n \geq 1$.

The initial choices I_0 and T_0 satisfy $(F2)_0$ and $(F7)_0$ by (6.7) and (6.10). Suppose that $I_0, \dots, I_n, T_0, \dots, T_n$, and, when $n \geq 1$, $r_0, \dots, r_{n-1}$ have been constructed and satisfy all applicable induction hypotheses. We choose $Q_n, r_n, T_{n+1}, I_{n+1}$ in this order and verify the required conditions at the next stage.

Since $J_n \subseteq I_n$ and $(F2)_n$ gives $I_n \cap \mathcal{Z}_n = \emptyset$,

$$W_d(x) \neq 0 \quad (x \in J_n, \ 2 \leq d \leq D(A_n)).$$

Thus Proposition 5.1 applies on J_n with $A = A_n$, and $(F7)_n$ is precisely the additional hypothesis (6.1) of Lemma 6.1, with $H = T_n$.

Before choosing the source height, define

$$(6.14) \quad \Lambda_{n+1} := \#\mathcal{Z}_{n+1} + 2.$$

Notice that $\mathcal{Z}_{n+1}$, and hence Λ_{n+1} , is already fixed at this point, independently of all objects to be chosen at stage n .

Choose an integer Q_n sufficiently large that the conclusion of Lemma 6.1 holds and, simultaneously,

- (Q1) $Q_n > 2q_{n-1}$ if $n \geq 1$;

- (Q2)

$$(2Q_n)^{-A_n} < \frac{|I_n|}{3};$$

- (Q3)

$$Q_n^{A_n/97} \geq 2T_n + 2;$$

- (Q4)

$$(6.15) \quad 3C_{\text{low}} \Lambda_{n+1} 2^{A_n+\tau-2} Q_n^{-A_n/97} \leq \frac{c_F}{4}.$$

Such a choice is possible. Indeed, at this stage $J_n, A_n, T_n, \Lambda_{n+1}$, and, when $n \geq 1$, q_{n-1} are fixed, as are all constants and thresholds in Lemma 6.1. Conditions (Q1)–(Q3) impose only finitely many lower bounds on Q_n , while the left-hand side of (6.15) tends to zero as $Q_n \rightarrow \infty$.

Choose a reduced safe fraction

$$(6.16) \quad r_n = \frac{p_n}{q_n} \in J_n, \quad Q_n \leq q_n < 2Q_n.$$

Set

$$(6.17) \quad R_n := (2Q_n)^{-A_n}, \quad T_{n+1} := \left\lfloor Q_n^{A_n/97} \right\rfloor,$$

and define

$$(6.18) \quad K_{n+1} := (r_n - R_n, r_n + R_n).$$

Condition (Q3) gives $T_{n+1} \geq 2T_n + 2 > T_n$, proving (F3) $_n$. Since $r_n \in J_n$ and $\text{dist}(J_n, \mathbb{R} \setminus I_n) = |I_n|/3$, condition (Q2) gives

$$K_{n+1} \subset \text{int}(I_n).$$

We transfer the safety of r_n to K_{n+1} . If $T_n \leq b < T_{n+1}$, then

$$b < T_{n+1} \leq Q_n^{A_n/97} = T(Q_n, A_n).$$

Therefore the safety of r_n gives, for every $a \in \mathbb{Z}$,

$$(6.19) \quad |f(r_n) - a/b| > 2b^{-\tau} + 4M_*Q_n^{-A_n}.$$

For $x \in K_{n+1}$, the mean-value theorem and (5.4) yield

$$|f(x) - f(r_n)| \leq M_* |x - r_n| < M_* R_n \leq M_* Q_n^{-A_n}.$$

Consequently,

$$(6.20) \quad |f(x) - a/b| > 2b^{-\tau} + 3M_*Q_n^{-A_n} > b^{-\tau}.$$

It remains to choose I_{n+1} inside K_{n+1} . For every $k < n$, property (F5) $_k$ gives $r_k \notin I_{k+1}$, and repeated use of (F1) gives $I_n \subseteq I_{k+1}$. Hence no previously chosen center r_k , $k < n$, lies in K_{n+1} . It is therefore enough to remove

$$(6.21) \quad F_{n+1} := \mathcal{Z}_{n+1} \cup \{r_n\}.$$

The set $F_{n+1} \cap K_{n+1}$ has at most $\#\mathcal{Z}_{n+1} + 1 = \Lambda_{n+1} - 1$ points. Hence $K_{n+1} \setminus F_{n+1}$ has at most Λ_{n+1} connected components. Their total length is $2R_n$, so one component, say (u_{n+1}, v_{n+1}) , has length at least $2R_n/\Lambda_{n+1}$. Put

$$s_n := \frac{R_n}{\Lambda_{n+1}}$$

and choose

$$(6.22) \quad I_{n+1} := \left[u_{n+1} + \frac{s_n}{2}, u_{n+1} + \frac{3s_n}{2} \right].$$

Then $I_{n+1} \subseteq (u_{n+1}, v_{n+1})$ and

$$(6.23) \quad |I_{n+1}| = s_n = \frac{R_n}{\Lambda_{n+1}} = \frac{(2Q_n)^{-A_n}}{\Lambda_{n+1}}.$$

Thus $(F1)_n$ and $(F2)_{n+1}$ hold. Furthermore, $r_n \notin I_{n+1}$ and $I_{n+1} \subset K_{n+1}$, so for every $x \in I_{n+1}$,

$$0 < |x - r_n| < R_n \leq q_n^{-A_n},$$

where the last inequality follows from $q_n < 2Q_n$. This proves $(F5)_n$. Property $(F6)_n$ follows from (6.20). If $n \geq 1$, then $q_n \geq Q_n > 2q_{n-1}$ by (Q1), proving $(F4)_n$.

It remains to verify $(F7)_{n+1}$. From (6.23) and (6.9),

$$(6.24) \quad |J_{n+1}| = \frac{(2Q_n)^{-A_n}}{3\Lambda_{n+1}}.$$

Condition (Q3) implies $Q_n^{A_n/97} \geq 2$. Since $\lfloor X \rfloor \geq X/2$ for $X \geq 2$,

$$T_{n+1} \geq \frac{1}{2} Q_n^{A_n/97}.$$

Because $2 - \tau < 0$,

$$(6.25) \quad T_{n+1}^{2-\tau} \leq 2^{\tau-2} Q_n^{-A_n(\tau-2)/97}.$$

Combining (6.24) and (6.25) gives

$$(6.26) \quad \begin{aligned} \frac{C_{\text{low}} T_{n+1}^{2-\tau}}{|J_{n+1}|} &\leq 3C_{\text{low}} \Lambda_{n+1} 2^{A_n+\tau-2} Q_n^{A_n-A_n(\tau-2)/97} \\ &= 3C_{\text{low}} \Lambda_{n+1} 2^{A_n+\tau-2} Q_n^{-A_n/97} \\ &\leq \frac{c_F}{4}, \end{aligned}$$

where we used $97 = \tau - 3$ and (Q4). The negative exponent $-A_n/97$ ensures that the contribution of the next lower target cutoff becomes small relative to the length of the next source interval. This is $(F7)_{n+1}$. We have therefore completed the inductive step, and hence the construction.

By (F1), the intervals I_n are nested, nonempty, and compact. Moreover, for $n \geq 1$,

$$Q_n > 2q_{n-1} \geq 2Q_{n-1},$$

so $Q_n \rightarrow \infty$. Since $I_{n+1} \subset K_{n+1}$,

$$\text{diam}(I_{n+1}) \leq 2R_n = 2(2Q_n)^{-A_n} \longrightarrow 0.$$

The nested-interval theorem therefore gives a unique point

$$(6.27) \quad \{\xi\} = \bigcap_{n=0}^{\infty} I_n.$$

Property (F5) yields

$$(6.28) \quad 0 < |\xi - r_n| < q_n^{-A_n} \quad (n \geq 0).$$

The denominators q_n tend to infinity. Given $M \geq 1$, we have $A_n = n+3 \geq M$ for all sufficiently large n , and hence

$$0 < |\xi - r_n| < q_n^{-M}$$

for infinitely many n . Thus $\xi \in \mathcal{L}$.

By (F3), the positive integers T_n are strictly increasing and tend to infinity. The half-open blocks $[T_n, T_{n+1}) \cap \mathbb{Z}_{>0}$ therefore partition all integers $b \geq T_0$. For such a b , let n be the unique index with $T_n \leq b < T_{n+1}$. Since $\xi \in I_{n+1}$, property (F6) gives, for every $a \in \mathbb{Z}$,

$$(6.29) \quad |f(\xi) - a/b| > b^{-\tau} = b^{-100}.$$

This is the asserted eventual lower bound.

For every $\lambda > \tau$, the inequality $0 < |f(\xi) - a/b| < b^{-\lambda}$ can hold only for $b < T_0$, and for each such b there are only finitely many possible numerators. Hence $\mu(f(\xi)) \leq \tau = 100$. Finally, $\xi \in I_0 \subseteq I_* \subseteq V$. $\square$

Remark 6.2. The escape set

$$\mathcal{E}_\tau(f) := \{\xi \in \mathcal{L} \cap U : \mu(f(\xi)) \leq \tau\}$$

contains a Cantor subset in every nonempty open subinterval of U . Indeed, the component (u_{n+1}, v_{n+1}) used in the inductive step has length at least $2R_n/\Lambda_{n+1}$. Choose inside it two disjoint compact intervals, each of length $R_n/(2\Lambda_{n+1})$. Both inherit the approximation and exclusion estimates, and (F7) holds for both after replacing the factor 3 in (Q4) by 6. Repeating this choice at every node gives a nested binary construction whose interval diameters tend to zero. Every branch yields a point of the escape set, and the resulting set is a Cantor set.

6.3. Proof of Corollary 1.3. If the conclusion were false, then f would not be the restriction to U of any rational function in $\mathbb{R}(x)$. Theorem 1.2 would then produce a point $\xi \in \mathcal{L} \cap U$ for which $\mu(f(\xi)) \leq 100$. Hence $f(\xi) \notin \mathcal{L}$, contradicting $f(\mathcal{L} \cap U) \subseteq \mathcal{L}$. $\square$

6.4. Proof of Theorem 1.1. Let $F : \mathbb{C} \rightarrow \mathbb{C}$ be entire and suppose that $F(\mathcal{L}) \subseteq \mathcal{L}$. Since $\mathcal{L}$ is dense in $\mathbb{R}$, every $x \in \mathbb{R}$ is the limit of a sequence (ξ_n) in $\mathcal{L}$. The values $F(\xi_n)$ are real, and continuity gives

$$F(x) = \lim_{n \rightarrow \infty} F(\xi_n) \in \mathbb{R}.$$

Thus $F(\mathbb{R}) \subseteq \mathbb{R}$.

Apply Corollary 1.3 to the real-analytic function $F|_{\mathbb{R}}$. There are polynomials $P, Q \in \mathbb{R}[z]$, with $Q \neq 0$ and with no common zero in $\mathbb{C}$, such that

$$F(x) = \frac{P(x)}{Q(x)} \quad (x \in \mathbb{R}).$$

The entire function

$$z \mapsto Q(z)F(z) - P(z)$$

vanishes on $\mathbb{R}$ and hence vanishes identically on $\mathbb{C}$. If Q were nonconstant, let $\zeta \in \mathbb{C}$ be a zero of Q . The identity $QF - P \equiv 0$ would give $P(\zeta) = 0$, contrary to the coprimality of P and Q . Hence Q is constant, and therefore $F \in \mathbb{R}[z]$. $\square$

7. ARITHMETIC RIGIDITY AND NONLINEAR SHARPNESS

Corollary 1.3 leaves open the coefficient field of rational functions with Maillet's property. We show that a Möbius transformation has this property exactly when it belongs to $\mathbb{Q}(x)$, whereas in every higher degree there are preserving polynomials with transcendental coefficients.

7.1. Möbius rigidity. We prove Theorem 1.4. Let

$$(7.1) \quad M(x) = \frac{ax + b}{cx + d}, \quad \Delta_M := ad - bc \neq 0,$$

with real coefficients. Since multiplying a, b, c, d by a common nonzero scalar does not change M , the coefficient vector naturally determines a point

$$[a : b : c : d] \in \mathbb{P}^3(\mathbb{R}),$$

and

$$M \in \mathbb{Q}(x) \iff [a : b : c : d] \in \mathbb{P}^3(\mathbb{Q}).$$

Choose an index $\ell \in \{1, 2, 3, 4\}$ for which the corresponding coordinate of

$$\mathbf{u} := (a, b, c, d)$$

is nonzero, and rescale $\mathbf{u}$ so that its ℓ -th coordinate is equal to 1. We continue to denote the normalized coefficients by a, b, c, d , and put $\Delta_M = ad - bc$. Let $\boldsymbol{\theta} \in \mathbb{R}^3$ be the vector of the remaining coordinates, in their inherited order. Then

$$(7.2) \quad M \in \mathbb{Q}(x) \iff \boldsymbol{\theta} \in \mathbb{Q}^3.$$

For $h \in \mathbb{Z}_{>0}$, define

$$(7.3) \quad \Delta_h(\boldsymbol{\theta}) := \min_{\mathbf{r} \in \mathbb{Z}^3} \left\| \boldsymbol{\theta} - \frac{\mathbf{r}}{h} \right\|_{\infty}.$$

By (7.2), if $M \notin \mathbb{Q}(x)$, then $\Delta_h(\boldsymbol{\theta}) > 0$ for every $h \geq 1$.

The proof splits according to the approximation properties of this coefficient vector. Arbitrarily strong rational approximations lead to a Liouville point in a rational fiber. Otherwise, a Diophantine lower bound, combined with three-point reconstruction, yields a two-height counting estimate.

Lemma 7.1 (Simultaneous approximation dichotomy). *Assume that $M \notin \mathbb{Q}(x)$. Exactly one of the following alternatives holds:*

(a) *for every $N, H \in \mathbb{Z}_{>0}$ there exists $h \geq H$ such that*

$$\Delta_h(\boldsymbol{\theta}) < h^{-N};$$

(b) *there exist an integer $\nu \geq 1$ and a constant $\eta_0 > 0$ such that*

$$(7.4) \quad \left\| \boldsymbol{\theta} - \frac{\mathbf{r}}{h} \right\|_{\infty} \geq \eta_0 h^{-\nu}$$

for every $h \in \mathbb{Z}_{>0}$ and every $\mathbf{r} \in \mathbb{Z}^3$.

Proof. If (a) fails, there exist $N, H \in \mathbb{Z}_{>0}$ such that $\Delta_h(\boldsymbol{\theta}) \geq h^{-N}$ for every $h \geq H$. Increasing H if necessary, we may assume that $H \geq 2$. Since the finitely many numbers $\Delta_h(\boldsymbol{\theta})$, $1 \leq h < H$, are positive, one may take $\nu = N$ and

$$\eta_0 := \min \left\{ 1, \min_{1 \leq h < H} h^N \Delta_h(\boldsymbol{\theta}) \right\} > 0.$$

This gives (b). Conversely, (b) plainly excludes (a). $\square$

In alternative (a), rational approximation of the coefficient vector transfers to rational approximation of the inverse image of a fixed rational value.

Lemma 7.2 (Rational fibers in the projectively Liouville case). *Assume that $M \notin \mathbb{Q}(x)$ and that alternative (a) of Lemma 7.1 holds. Then every nonempty open interval V on which M is defined contains a point $\xi \in \mathcal{L}$ such that*

$$M(\xi) \in \mathbb{Q}.$$

Proof. The image $M(V)$ is a nonempty open interval. Choose $\rho \in \mathbb{Q} \cap M(V)$ such that

$$(7.5) \quad \xi := M^{-1}(\rho)$$

is irrational. Such a choice is possible. Indeed, if three distinct rational values had rational preimages, then M would send three distinct rational points to three distinct rational points. A Möbius transformation is uniquely determined by its values at three distinct points, and the corresponding linear system has rational coefficients; hence its projective coefficient vector would belong to $\mathbb{P}^3(\mathbb{Q})$, contrary to (7.2). By construction, $\xi \in V$.

Using alternative (a) recursively, choose strictly increasing integers h_n and vectors $\mathbf{r}_n \in \mathbb{Z}^3$ such that

$$(7.6) \quad \left\| \boldsymbol{\theta} - \frac{\mathbf{r}_n}{h_n} \right\|_{\infty} < h_n^{-n} \quad (n \geq 1).$$

Insert the three coordinates of $\mathbf{r}_n/h_n$ into the fixed chart, keeping the ℓ -th coordinate equal to 1, and write the resulting coefficient vector as

$$\mathbf{u}_n = (a^{(n)}, b^{(n)}, c^{(n)}, d^{(n)}).$$

Since $\mathbf{u}_n \rightarrow \mathbf{u}$, we have $a^{(n)}d^{(n)} - b^{(n)}c^{(n)} \neq 0$ for all sufficiently large n . Let $M_n \in \mathbb{Q}(x)$ be the corresponding Möbius transformation. Since $a - \rho c \neq 0$, the map sending a coefficient vector near $\mathbf{u}$ to the finite preimage of ρ is smooth. Hence

$$\xi_n := M_n^{-1}(\rho) \in \mathbb{Q}, \quad |\xi - \xi_n| \leq C_1 h_n^{-n}$$

for some $C_1 > 0$ and all sufficiently large n .

After multiplying $\mathbf{u}_n$ by h_n , all four coefficients are integers of size $O(h_n)$. Writing $\rho = r/s$ in lowest terms and using

$$M_n^{-1}(\rho) = \frac{\rho d^{(n)} - b^{(n)}}{a^{(n)} - \rho c^{(n)}},$$

we see that the positive denominator q'_n of ξ_n in lowest terms satisfies

$$(7.7) \quad q'_n \leq C_2 h_n$$

for some $C_2 > 0$. Moreover, $\xi_n \rightarrow \xi$, and ξ is irrational, so $q'_n \rightarrow \infty$.

Fix $A \geq 1$. For all sufficiently large $n > A$,

$$0 < |\xi - \xi_n| \leq C_1 h_n^{-n} < C_2^{-A} h_n^{-A} \leq (q'_n)^{-A}.$$

Thus $\xi \in \mathcal{L}$, while $M(\xi) = \rho \in \mathbb{Q}$. $\square$

We now turn to the Diophantine case. The following reconstruction lemma replaces the determinant step in Section 5: three rational points sufficiently close to the graph of M determine a rational Möbius transformation with nearby coefficients and controlled height.

Lemma 7.3 (Three-point rational reconstruction). *Fix the Möbius transformation M in (7.1), with the normalization above, and a nondegenerate compact interval I_* on which M is defined. There exist constants $c_0, C_0 > 0$, depending only on M and I_* , with the following property.*

Let $x_1, x_2, x_3 \in I_$ be distinct, let $y_1, y_2, y_3 \in \mathbb{R}$, and put*

$$\varepsilon := \max_{1 \leq i \leq 3} |y_i - M(x_i)|, \quad V(x_1, x_2, x_3) := \prod_{1 \leq i < j \leq 3} |x_i - x_j|.$$

If

$$(7.8) \quad \varepsilon \leq c_0 V(x_1, x_2, x_3),$$

then there is a coefficient vector $\hat{\mathbf{u}}$, normalized in the same chart as $\mathbf{u}$, such that

$$(7.9) \quad \|\hat{\mathbf{u}} - \mathbf{u}\|_\infty \leq C_0 \frac{\varepsilon}{V(x_1, x_2, x_3)}.$$

It represents a nonconstant Möbius transformation $\widehat{M}$ satisfying $\widehat{M}(x_i) = y_i$ for $1 \leq i \leq 3$.

If, in addition, for some positive real numbers Q, B ,

$$x_i = \frac{p_i}{q_i}, \quad y_i = \frac{r_i}{b_i}, \quad q_i < 2Q, \quad b_i < 2B,$$

with $p_i, r_i \in \mathbb{Z}$ and $q_i, b_i \in \mathbb{Z}_{>0}$, then the three free coordinates of $\hat{\mathbf{u}}$ can be written with a common positive denominator h satisfying

$$(7.10) \quad 1 \leq h \leq C_0 Q^3 B^3.$$

Proof. For $x, y \in \mathbb{R}$, put

$$v(x, y) := (x, 1, -xy, -y) \in \mathbb{R}^4.$$

Let $\mathcal{A}_0$ be the 3×4 matrix with rows $v(x_i, M(x_i))$, and let $\mathcal{A}$ have rows $v(x_i, y_i)$. For a 3×4 matrix $\mathcal{B}$, let $w_j(\mathcal{B}) := (-1)^{j+1} \det \mathcal{B}^{(j)}$, where $\mathcal{B}^{(j)}$ is obtained by deleting column j . Then $\mathcal{B}\mathbf{w}(\mathcal{B})^\top = 0$. A direct calculation gives

$$(7.11) \quad \mathbf{w}(\mathcal{A}_0) = \Lambda(x_1, x_2, x_3)\mathbf{u},$$

where

$$\Lambda(x_1, x_2, x_3) = \frac{\Delta_M(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)}{(cx_1 + d)(cx_2 + d)(cx_3 + d)}.$$

Since I_* is compact and contains no pole of M ,

$$(7.12) \quad |\Lambda(x_1, x_2, x_3)| \geq c_1 V(x_1, x_2, x_3)$$

for some $c_1 > 0$.

First restrict $c_0 \leq 1$. All entries of $\mathcal{A}_0$ and, under (7.8), of $\mathcal{A}$ then lie in a fixed compact set. The maximal minors, being polynomial in the matrix entries, satisfy

$$(7.13) \quad \|\mathbf{w}(\mathcal{A}) - \mathbf{w}(\mathcal{A}_0)\|_\infty \leq C_1 \varepsilon,$$

with C_1 independent of any subsequent decrease in c_0 . Choose c_0 so small that the right-hand side is at most $|\Lambda|/2$. Since $u_\ell = 1$, we have $w_\ell(\mathcal{A}_0) = \Lambda$, and hence $|w_\ell(\mathcal{A})| \geq |\Lambda|/2 > 0$. Define

$$\hat{\mathbf{u}} := \frac{\mathbf{w}(\mathcal{A})}{w_\ell(\mathcal{A})}.$$

Equations (7.11)–(7.13) give

$$\|\hat{\mathbf{u}} - \mathbf{u}\|_\infty \leq \frac{2C_1(1 + \|\mathbf{u}\|_\infty)\varepsilon}{|\Lambda|},$$

which proves (7.9). Since $ad - bc \neq 0$ is an open condition on the coefficient vector, decreasing c_0 if necessary ensures that $\hat{\mathbf{u}}$ represents a nonconstant Möbius transformation. Its denominator cannot vanish at any x_i , since the kernel relation would then force its numerator to vanish there as well, contradicting its nonzero determinant. Thus the kernel relation gives $\widehat{M}(x_i) = y_i$.

For the height assertion, multiply the i -th row of $\mathcal{A}$ by $q_i b_i$. The resulting integer row is

$$(p_i b_i, q_i b_i, -p_i r_i, -q_i r_i).$$

The numbers x_i lie in I_* , and (7.8) bounds the y_i uniformly in terms of M and I_* . Thus every entry of this integer matrix is $O_{M, I_*}(QB)$. Its cofactor vector $\mathbf{k} \in \mathbb{Z}^4$ spans the same one-dimensional kernel as $\mathbf{w}(\mathcal{A})$, and each of its coordinates is $O_{M, I_*}(Q^3 B^3)$. Since $k_\ell \neq 0$, the normalized coordinates are k_j/k_ℓ ; taking $h = |k_\ell|$ proves (7.10), after enlarging C_0 if necessary. No reduction of this common denominator is needed. $\square$

For the Diophantine branch, assume that alternative (b) of Lemma 7.1 holds, and fix ν, η_0 as in (7.4) for this M . Fix a nondegenerate compact interval I_* on which M is defined. Let m_* and M_* be the minimum and maximum of $|M'|$ on I_* , so that

$$(7.14) \quad 0 < m_* \leq |M'(x)| \leq M_* \quad (x \in I_*).$$

Choose an integer

$$(7.15) \quad \tau_1 \geq 9\nu + 18$$

and, for integers $A \geq 3$ and $Q \geq 2$, put

$$(7.16) \quad T_1(Q, A) := Q^{A/(\tau_1-3)}.$$

For a nondegenerate compact interval $J \subseteq I_*$ and integers $H, Q \geq 2$, define

$$(7.17) \quad \begin{aligned} \mathcal{D}_M(J; Q, A, H) := & \left\{ \frac{p}{q} \in J : \gcd(p, q) = 1, \quad Q \leq q < 2Q, \right. \\ & \exists r \in \mathbb{Z}, b \in \mathbb{Z}_{>0} \text{ with } H \leq b < T_1(Q, A), \\ & \left. |M(p/q) - r/b| \leq 2b^{-\tau_1} + 4M_*Q^{-A} \right\}. \end{aligned}$$

As in Section 5, the source fractions are reduced, whereas the target fractions need not be.

Proposition 7.4 (Möbius two-height counting estimate). *There exists a constant $C_{\text{mob}} > 0$, depending only on M, I_* , and τ_1 , with the following property. For every integer $A \geq 3$, there exist constants $C_A > 0$ and $Q_A \in \mathbb{Z}_{>0}$ such that*

$$(7.18) \quad \#\mathcal{D}_M(J; Q, A, H) \leq C_{\text{mob}}Q^2H^{2-\tau_1} + C_AQ$$

for every nondegenerate compact interval $J \subseteq I_*$, every integer $H \geq 2$, and every integer $Q \geq Q_A$. The constants C_A and Q_A are uniform in J and H .

Proof. Fix $A \geq 3$. All implied constants below may depend on the fixed data $M, I_*, \nu, \eta_0, \tau_1$, but not on J, H, Q or the target block; any additional dependence on A is indicated. If $T_1(Q, A) \leq H$, then

$$\mathcal{D}_M(J; Q, A, H) = \emptyset.$$

Otherwise, decompose $[H, T_1(Q, A)) \cap \mathbb{Z}_{>0}$ into the dyadic blocks

$$\mathcal{B}_k := [B_k, \min\{2B_k, T_1(Q, A)\}) \cap \mathbb{Z}_{>0}, \quad B_k := 2^k H,$$

with $B_k < T_1(Q, A)$, and let $\mathcal{D}_k$ be the source centers admitting a witness with denominator in $\mathcal{B}_k$. Then $\#\mathcal{D}_M(J; Q, A, H) \leq \sum_k \#\mathcal{D}_k$. We call a block small if $B_k < Q^{1/2}$ and large otherwise.

Small target heights. For a small block with left endpoint B , the approximation condition and the boundedness of M on I_* show that there are $O(B^2)$

pairs (r, b) that can occur as witnesses. For each such pair, monotonicity and (7.14) show that

$$E_{r,b} := \left\{ x \in J : |M(x) - r/b| \leq 2b^{-\tau_1} + 4M_*Q^{-A} \right\}$$

is empty, a singleton, or an interval of length

$$|E_{r,b}| \ll b^{-\tau_1} + Q^{-A}.$$

Consequently, their union E_B has $O(B^2)$ components and

$$|E_B| \ll B^{2-\tau_1} + B^2Q^{-A}.$$

Lemma 2.1 gives

$$\#\mathcal{D}_k \ll Q^2B^{2-\tau_1} + Q^{2-A}B^2 + B^2.$$

Summing over $B = 2^kH < Q^{1/2}$, we obtain

$$(7.19) \quad \sum_{B_k < Q^{1/2}} \#\mathcal{D}_k \leq C_{\text{mob}}Q^2H^{2-\tau_1} + O_A(Q),$$

with C_{mob} independent of J, A, H, Q .

Large target heights. Fix a large block with left endpoint $B = Q^u$. Then

$$(7.20) \quad u \geq \frac{1}{2}, \quad u < \frac{A}{\tau_1 - 3}.$$

We shall show that, for all sufficiently large Q , this block contains at most two resonant centers. Indeed, three such centers would determine, via Lemma 7.3, a rational Möbius transformation sufficiently close to M to contradict the Diophantine barrier (7.4). Suppose therefore that the block contains three distinct resonant centers

$$x_i = \frac{p_i}{q_i}, \quad Q \leq q_i < 2Q, \quad 1 \leq i \leq 3,$$

with witnesses

$$y_i = \frac{r_i}{b_i}, \quad B \leq b_i < 2B.$$

Put

$$\delta := 2B^{-\tau_1} + 4M_*Q^{-A}, \quad s := \min\{\tau_1 u, A\}.$$

Then $|y_i - M(x_i)| \leq \delta \ll Q^{-s}$, and

$$(7.21) \quad s \geq (\tau_1 - 3)u.$$

Since the source fractions are distinct and reduced,

$$V(x_1, x_2, x_3) = \prod_{i < j} |x_i - x_j| > \frac{1}{64Q^6}.$$

The choice (7.15) implies $Q^6\delta \rightarrow 0$, uniformly in the block. Hence Lemma 7.3 applies for all sufficiently large Q , and produces a rational coefficient vector in the fixed chart satisfying

$$(7.22) \quad \|\hat{\mathbf{u}} - \mathbf{u}\|_\infty \ll Q^6\delta \ll Q^{6-s}.$$

Its free coordinates have a common positive denominator h with

$$1 \leq h \ll Q^3 B^3 = Q^{3(1+u)}.$$

Applying (7.4) to these coordinates gives

$$(7.23) \quad \|\hat{\mathbf{u}} - \mathbf{u}\|_\infty \gg Q^{-3\nu(1+u)}.$$

The exponent gap between (7.22) and (7.23) satisfies

$$(7.24) \quad \begin{aligned} s - 6 - 3\nu(1+u) &\geq (\tau_1 - 3 - 3\nu)u - (3\nu + 6) \\ &\geq \frac{\tau_1 - 3 - 3\nu}{2} - (3\nu + 6) \geq \frac{3}{2}, \end{aligned}$$

where we used $u \geq 1/2$ and (7.15). Thus the reconstructed coefficient vector would violate the Diophantine barrier for all sufficiently large Q , uniformly in the block, in J , and in H . Each large block therefore contains at most two resonant centers; no subdivision into short source cells is needed.

There are at most

$$1 + \log_2 \frac{T_1(Q, A)}{H} \leq 1 + \frac{A}{\tau_1 - 3} \log_2 Q$$

large blocks, so their total contribution is $O_A(\log Q) = O_A(Q)$. Together with (7.19), this proves (7.18). Every lower threshold imposed on Q depends only on the fixed data and A , not on J or H . $\square$

The preceding counting estimate supplies safe rational centers at every sufficiently large source scale. A nested-interval construction now turns these centers into the required Liouville point.

Lemma 7.5 (Finite-exponent escape in the Diophantine case). *Assume that alternative (b) of Lemma 7.1 holds. Then every nonempty open interval V on which M is defined contains a point $\xi \in \mathcal{L}$ such that*

$$\mu(M(\xi)) \leq \tau_1.$$

Proof. We follow the fusion argument of Section 6, with Proposition 7.4 supplying the counting estimate. There are no Wronskian zero sets to avoid, so the choice of the next interval is simpler.

Choose a nondegenerate compact interval $I_0 \Subset V$, and use it as I_* in Proposition 7.4. Write $J(I)$ for the middle third of I , as in Section 6. We construct nested nondegenerate compact intervals I_n , strictly increasing positive integers T_n , and reduced fractions $r_n = p_n/q_n$.

Put $A_n := n + 3$ and $J_n := J(I_n)$. Choose $T_0 \geq 2$ so large that

$$(7.25) \quad C_{\text{mob}} T_0^{2-\tau_1} \leq \frac{c_F}{4} |J_0|.$$

Suppose that I_n and T_n have been chosen and satisfy

$$(7.26) \quad C_{\text{mob}} T_n^{2-\tau_1} \leq \frac{c_F}{4} |J_n|.$$

Proposition 7.4, Lemma 2.2, and (7.26) imply that every sufficiently large Q admits a reduced fraction

$$r = \frac{p}{q} \in J_n, \quad Q \leq q < 2Q,$$

satisfying

$$(7.27) \quad |M(r) - a/b| > 2b^{-\tau_1} + 4M_*Q^{-A_n}$$

for every $a \in \mathbb{Z}$ and every integer b with $T_n \leq b < T_1(Q, A_n)$.

Choose an integer $Q_n \geq 2$ large enough for this conclusion and, simultaneously,

$$(7.28) \quad Q_n > 2q_{n-1} \quad (n \geq 1),$$

$$(7.29) \quad (2Q_n)^{-A_n} < \frac{|I_n|}{3},$$

$$(7.30) \quad Q_n^{A_n/(\tau_1-3)} \geq 2T_n + 2,$$

$$(7.31) \quad 12C_{\text{mob}}2^{A_n+\tau_1-2}Q_n^{-A_n/(\tau_1-3)} \leq \frac{c_F}{4}.$$

All quantities other than Q_n have already been fixed, and the left-hand side of (7.31) tends to zero. Select a safe center

$$r_n = \frac{p_n}{q_n} \in J_n, \quad Q_n \leq q_n < 2Q_n,$$

and put

$$R_n := (2Q_n)^{-A_n}, \quad T_{n+1} := \left\lfloor Q_n^{A_n/(\tau_1-3)} \right\rfloor.$$

Condition (7.30) gives $T_{n+1} > T_n$. Define

$$(7.32) \quad I_{n+1} := \left[r_n + \frac{R_n}{4}, r_n + \frac{R_n}{2} \right].$$

Since $r_n \in J_n$, condition (7.29) gives $I_{n+1} \subset \text{int}(I_n)$. For every $x \in I_{n+1}$,

$$(7.33) \quad 0 < |x - r_n| < R_n < q_n^{-A_n}.$$

Moreover, if $T_n \leq b < T_{n+1}$, then $b < T_1(Q_n, A_n)$, and (7.27) together with $|M(x) - M(r_n)| \leq M_*|x - r_n|$ gives

$$(7.34) \quad |M(x) - a/b| > b^{-\tau_1} \quad (a \in \mathbb{Z}, x \in I_{n+1}).$$

The interval I_{n+1} has length $R_n/4$, so $|J_{n+1}| = R_n/12$. Since $T_{n+1} \geq \frac{1}{2}Q_n^{A_n/(\tau_1-3)}$, condition (7.31) gives

$$\frac{C_{\text{mob}}T_{n+1}^{2-\tau_1}}{|J_{n+1}|} \leq 12C_{\text{mob}}2^{A_n+\tau_1-2}Q_n^{-A_n/(\tau_1-3)} \leq \frac{c_F}{4}.$$

This is (7.26) at stage $n+1$, completing the inductive step.

The nested intervals have diameters tending to zero, and hence $\bigcap_{n \geq 0} I_n = \{\xi\}$ for some $\xi \in V$. From (7.28) and (7.33), $q_n \rightarrow \infty$ and

$$0 < |\xi - r_n| < q_n^{-A_n}.$$

Since $A_n \rightarrow \infty$, we have $\xi \in \mathcal{L}$. Finally, the blocks $[T_n, T_{n+1}) \cap \mathbb{Z}_{>0}$ cover all sufficiently large denominators, and (7.34) gives

$$|M(\xi) - a/b| > b^{-\tau_1} \quad (a \in \mathbb{Z}, b \geq T_0).$$

Therefore $\mu(M(\xi)) \leq \tau_1$. $\square$

Together with Lemma 7.2, this treats both alternatives in Lemma 7.1 and yields the degree-one classification.

Proof of Theorem 1.4. If $M \in \mathbb{Q}(x)$, then Maillet's theorem gives Maillet's property. Conversely, suppose that $M \notin \mathbb{Q}(x)$. By Lemma 7.1, either alternative (a) or alternative (b) holds. In the first case, Lemma 7.2 produces, in every open interval on which M is defined, a Liouville number whose image is rational. In the second case, Lemma 7.5 produces a Liouville number whose image has finite irrationality exponent. In either case Maillet's property fails. This proves both assertions. $\square$

7.2. Nonlinear sharpness. Möbius rigidity cannot be extended to higher degree. Recall that a Liouville number σ , with continued-fraction convergents p_n/q_n , is called *strong Liouville* if

$$|\sigma - p_n/q_n| < q_n^{-\omega_n}$$

for some sequence $\omega_n \rightarrow \infty$. Equivalently (see [26]),

$$(7.35) \quad \frac{\log q_{n+1}}{\log q_n} \longrightarrow \infty.$$

We shall use the following consequence of Petruska's theorem [26, Theorem 1]: if σ is strong Liouville and $y \in \mathcal{L}$, then

$$(7.36) \quad \sigma y \in \mathbb{Q} \cup \mathcal{L}.$$

The rational alternative in (7.36) is essential. To exclude it for the monomials constructed below, we shall place their nonzero rational fibers in Mahler's classes U_m , with $m \geq 2$.

Here $h(\alpha)$ denotes the *absolute logarithmic Weil height* of an algebraic number α . We use the following sequential height formulation: a transcendental number κ belongs to U_m if m is the least positive integer for which there exist algebraic numbers β_n of degree m and positive real numbers w_n such that

$$h(\beta_n) \rightarrow \infty, \quad w_n \rightarrow \infty, \quad |\kappa - \beta_n| \leq e^{-w_n h(\beta_n)}.$$

The preceding formulation is the absolute-logarithmic-height version of LeVeque's definition of U_m -numbers [5, Remark 2.6] and, by the equivalence of Mahler's and Koksma's classifications [10, Theorem 3.6], defines the usual U_m -classes. In particular, $U_1 = \mathcal{L}$, and the classes U_m are pairwise disjoint. We shall use the following gap principle in the form of [4, Theorem 3.1 and Corollary 3.3].

Lemma 7.6 (U_m gap principle). *Let $m \geq 1$, let $\kappa \in \mathbb{C}$ be transcendental, and let (β_n) be algebraic numbers of degree at most m . Put $H_n := h(\beta_n)$, and let (w_n) be a sequence of positive real numbers. Suppose that*

$$H_n \rightarrow \infty, \quad w_n \rightarrow \infty, \quad |\kappa - \beta_n| \leq Ce^{-w_n H_n}, \quad H_{n+1} \leq Bw_n H_n$$

for fixed constants $B, C > 0$. If β_n has degree exactly m for all sufficiently large n , then $\kappa \in U_m$.

We now construct strong Liouville numbers for which the preceding gap principle applies simultaneously to every nonzero rational fiber and every degree $m \geq 2$. The denominator congruences enforce irreducibility in every degree, while rapid growth ensures strong Liouville approximation.

Theorem 7.7 (Nonlinear sharpness). *There exist continuum many positive strong Liouville numbers σ with the following property: for every integer $m \geq 2$ and every $r \in \mathbb{Q}^\times$, each root of*

$$(7.37) \quad X^m - \frac{r}{\sigma}$$

is a U_m -number. Consequently, for every such σ ,

$$(7.38) \quad \sigma \xi^m \in \mathcal{L} \quad (\xi \in \mathcal{L}, m \geq 2).$$

In particular, in every degree $m \geq 2$ there are Liouville-preserving polynomials in $\mathbb{R}[X] \setminus \mathbb{Q}[X]$.

Proof. Enumerate

$$\mathbb{Q}^\times = \{r_1, r_2, \dots\}, \quad r_j = \frac{A_j}{B_j},$$

where $A_j \in \mathbb{Z} \setminus \{0\}$, $B_j \in \mathbb{Z}_{>0}$, and $\gcd(A_j, B_j) = 1$. We construct

$$\sigma = [1; a_1, a_2, \dots], \quad \frac{p_n}{q_n} = [1; a_1, \dots, a_n],$$

with $q_{-1} = 0$ and $q_0 = 1$. Suppose that $a_1, \dots, a_{n-1}$ have been chosen. Select a prime ℓ_n such that

$$(7.39) \quad \ell_n \nmid q_{n-1} \prod_{j=1}^n A_j B_j.$$

Because q_{n-1} is invertible modulo ℓ_n^2 , one may choose a positive integer a_n , arbitrarily large, such that

$$(7.40) \quad q_n = a_n q_{n-1} + q_{n-2} \equiv \ell_n \pmod{\ell_n^2}$$

and

$$(7.41) \quad q_n > q_{n-1}^{n+1}.$$

Thus $\text{ord}_{\ell_n}(q_n) = 1$, while $\ell_n \nmid p_n$ because $\gcd(p_n, q_n) = 1$. Condition (7.41) implies (7.35); hence σ is strong Liouville, and $1 < \sigma < 2$.

Fix $j \geq 1$, $m \geq 2$, and write $\sigma_n = p_n/q_n$. For $n \geq j$, the polynomial

$$(7.42) \quad G_{j,m,n}(X) := B_j p_n X^m - A_j q_n$$

is Eisenstein at ℓ_n . Indeed, ℓ_n divides the constant coefficient exactly once and does not divide the leading coefficient; the intermediate coefficients vanish. Hence every root of

$$X^m - \frac{r_j}{\sigma_n}$$

has degree exactly m over $\mathbb{Q}$.

Fix a root $\beta \in \mathbb{C}$ satisfying $\beta^m = r_j/\sigma$. Since σ is transcendental, so is β . The holomorphic implicit-function theorem, applied to $zw^m - r_j = 0$ at (σ, β) , supplies a local branch $u(z)$ with $u(\sigma) = \beta$ and $u(z)^m = r_j/z$. For all sufficiently large n , put $\beta_n := u(\sigma_n)$. Then $[\mathbb{Q}(\beta_n) : \mathbb{Q}] = m$, and, since $1 < \sigma, \sigma_n < 2$,

$$(7.43) \quad |\beta - \beta_n| \ll_{j,m} |\sigma - \sigma_n| < \frac{C_{j,m}}{q_n q_{n+1}}.$$

Moreover,

$$m h(\beta_n) = h\left(\frac{A_j q_n}{B_j p_n}\right).$$

The greatest common divisor of $A_j q_n$ and $B_j p_n$ divides $|A_j B_j|$, while $1 < p_n/q_n < 2$. Therefore, with $H_n := h(\beta_n)$,

$$(7.44) \quad H_n = \frac{1}{m} \log q_n + O_{j,m}(1).$$

For sufficiently large n , we have $H_n > 0$; set

$$w_n := \frac{\log q_{n+1}}{2H_n}.$$

Then $w_n \rightarrow \infty$, and (7.43) gives

$$|\beta - \beta_n| \leq C_{j,m} e^{-w_n H_n}.$$

Since $m \geq 2$, equation (7.44) also gives

$$H_{n+1} \leq \log q_{n+1} = 2w_n H_n$$

for all sufficiently large n . After discarding finitely many terms, Lemma 7.6 therefore yields $\beta \in U_m$. Since j, m , and the chosen root were arbitrary, the asserted fiber property follows.

Now let $\xi \in \mathcal{L}$ and $m \geq 2$. Maillet's theorem gives $\xi^m \in \mathcal{L}$, and (7.36) gives

$$\sigma \xi^m \in \mathbb{Q} \cup \mathcal{L}.$$

If $\sigma \xi^m = r \in \mathbb{Q}$, then $r \neq 0$ and ξ is a root of $X^m - r/\sigma$; hence $\xi \in U_m$, contrary to $\xi \in U_1$. Thus (7.38) holds.

Finally, at every stage the prescribed congruence class modulo ℓ_n^2 contains infinitely many positive integers satisfying (7.41). Retaining two choices at each stage gives a binary tree of continuum many continued fractions, every one of which satisfies the preceding argument. $\square$

Proof of Theorem 1.5. Apply Theorem 7.7 and retain (7.38). $\square$

Theorems 1.4 and 7.7 show that coefficient rigidity has a sharp break between degree one and higher degree. They do not classify all rational preservers, but they show that the real coefficient field in Theorem 1.1 is unavoidable: the conclusion $F \in \mathbb{R}[z]$ cannot be strengthened to $F \in \overline{\mathbb{Q}}[z]$.

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OpenAI's ChatGPT was used solely for English-language editing and stylistic corrections. The author assumes full responsibility for the manuscript.

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